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Sholtis et al.

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(54) **SYSTEM AND METHOD TO FACILITATE AN ACCOUNT PROTECTION CHECK THROUGH BLOCKCHAIN**

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H04L 9/00 (2022.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04L 9/50** (2022.05); **H04L 9/30** (2013.01); **H04L 9/321** (2013.01)

(58) **Field of Classification Search**
CPC H04L 63/1416; H04L 63/1466; H04L 63/101; H04L 63/083; H04L 63/0263
See application file for complete search history.

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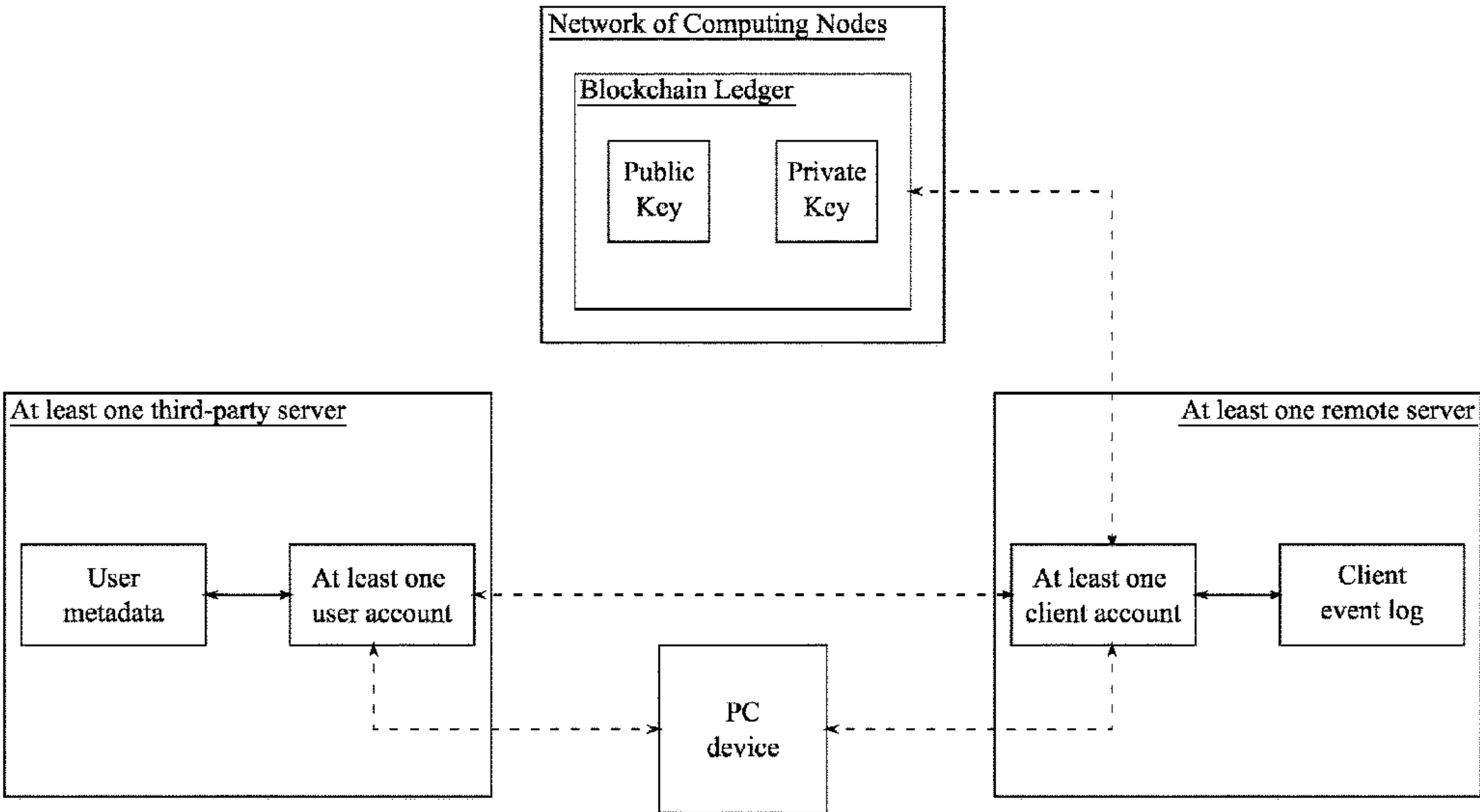
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Primary Examiner — Dao Q Ho

(57) **ABSTRACT**

A system and a method are provided for facilitating an account protection check through blockchain. The system and the method allow a service provider to utilize blockchain technology to validate the access request to various accounts. A public key and a private key are generated and recorded by a blockchain ledger that are used to access the accounts. A third-party server of a service provider requests a token from the system before the authentication process. If the credential set is in a locked status or unlocked status, at least one remote server of the system respectively relays an invalidation token or a validation token to the third-party server. If the invalidation token is relayed to the third-party server, the service provider does not go through the authentication process. If the validation token is relayed to the third-party server, the service provider proceeds with the authentication process as standard procedure.

19 Claims, 21 Drawing Sheets



Related U.S. Application Data

continuation-in-part of application No. 17/207,411,
filed on Mar. 19, 2021, now Pat. No. 11,615,180.

(60) Provisional application No. 62/991,990, filed on Mar.
19, 2020.

(51) **Int. Cl.**
H04L 9/30 (2006.01)
H04L 9/32 (2006.01)

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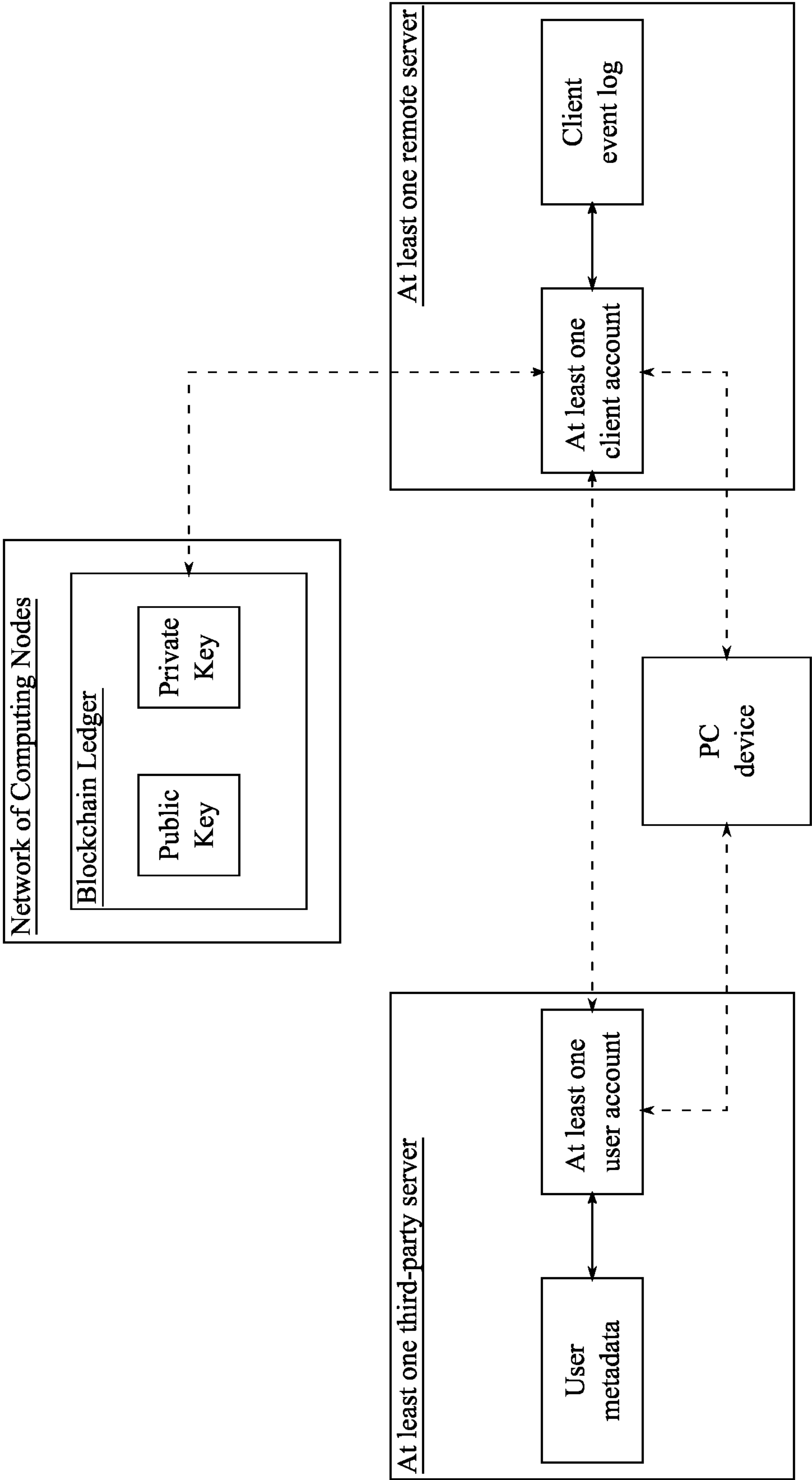


FIG. 1

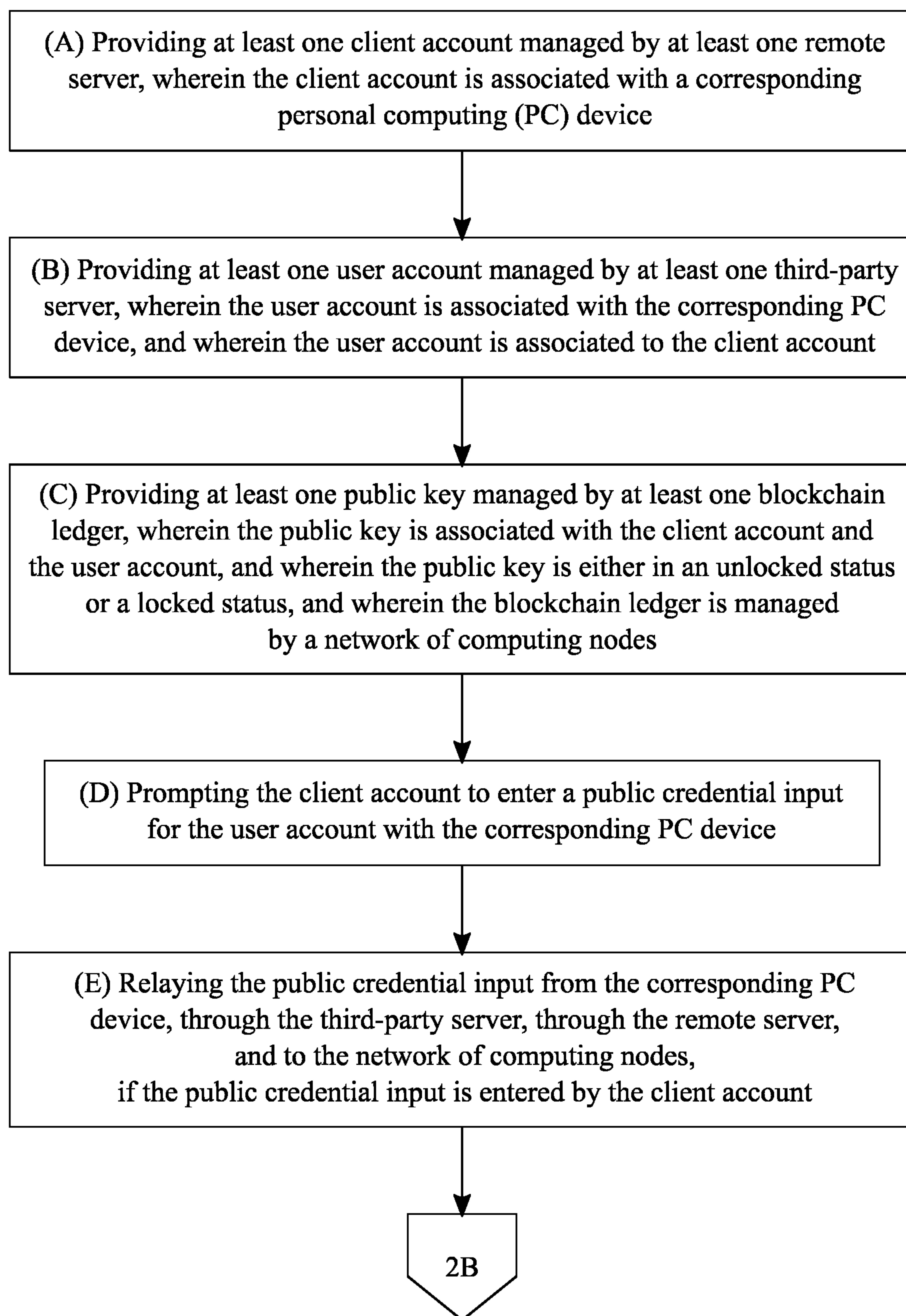


FIG. 2A

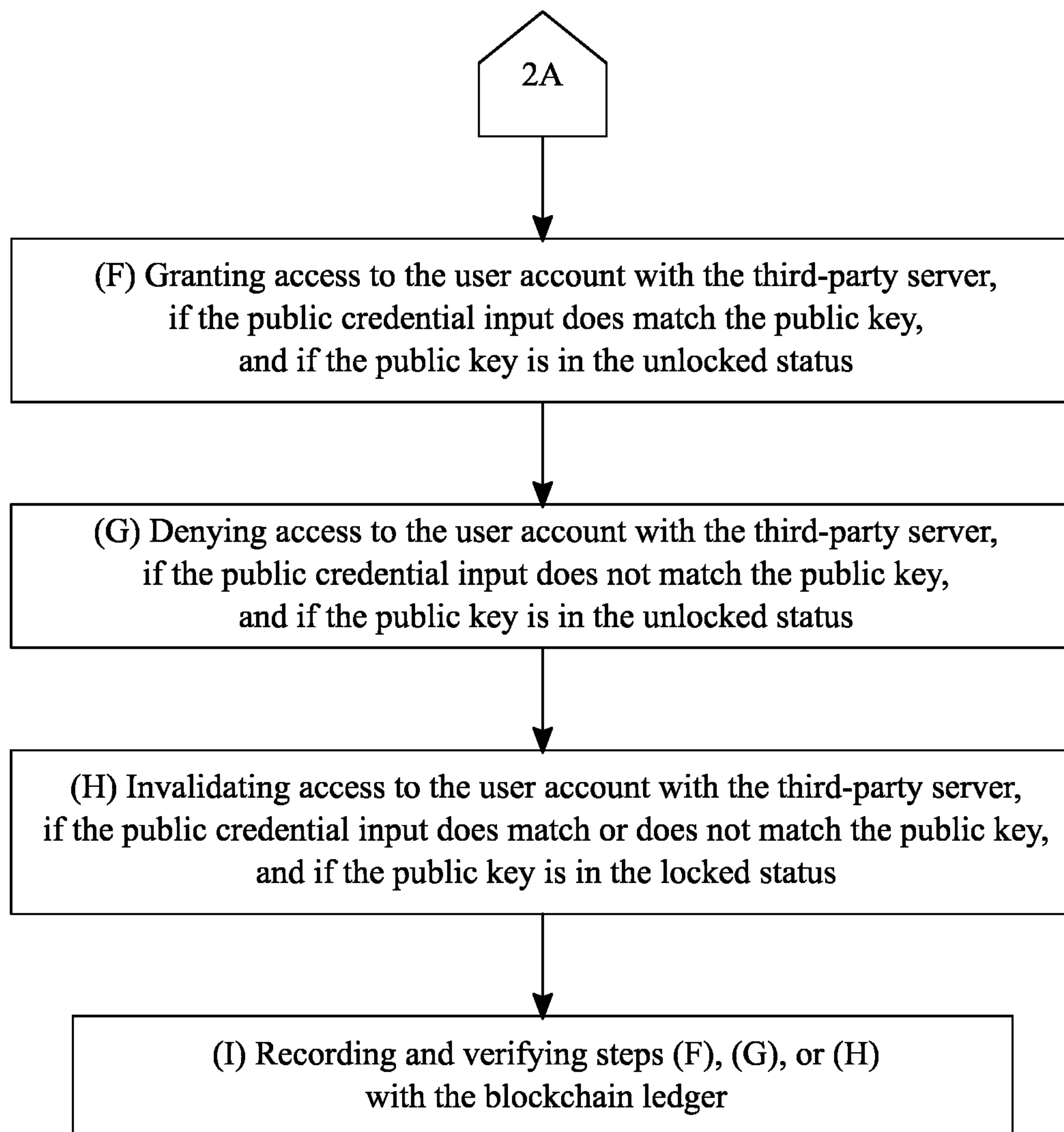


FIG. 2B

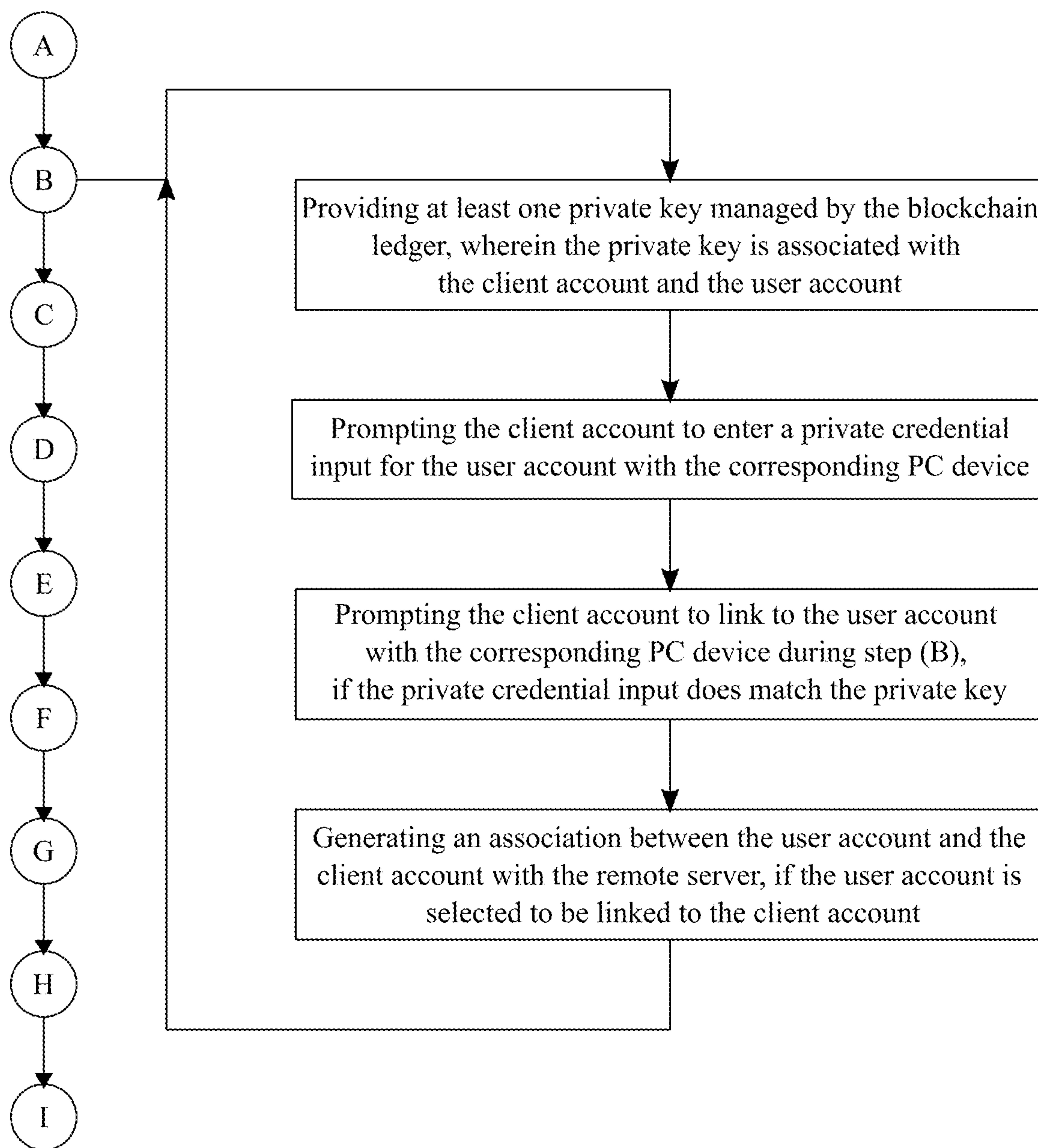


FIG. 3

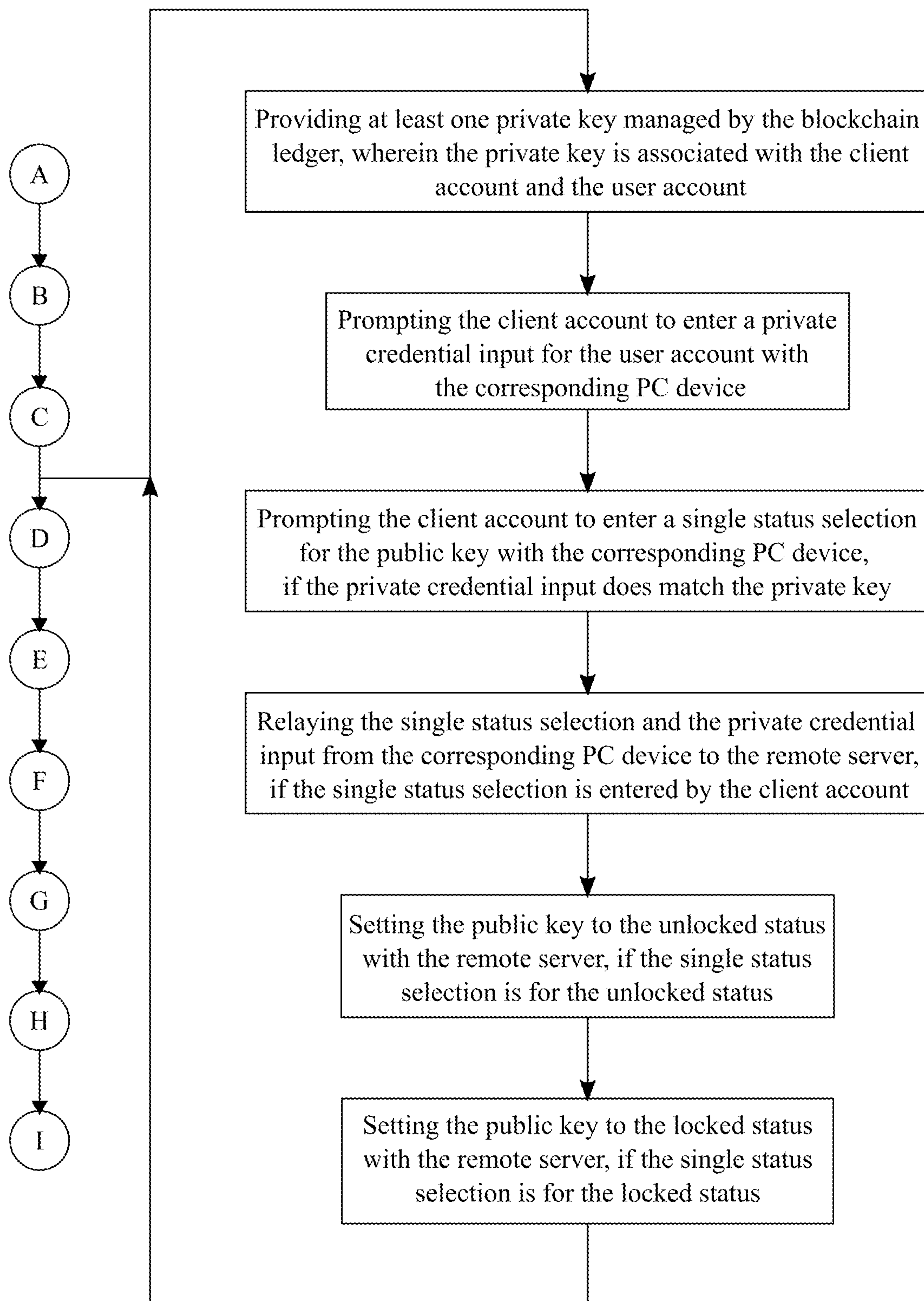


FIG. 4

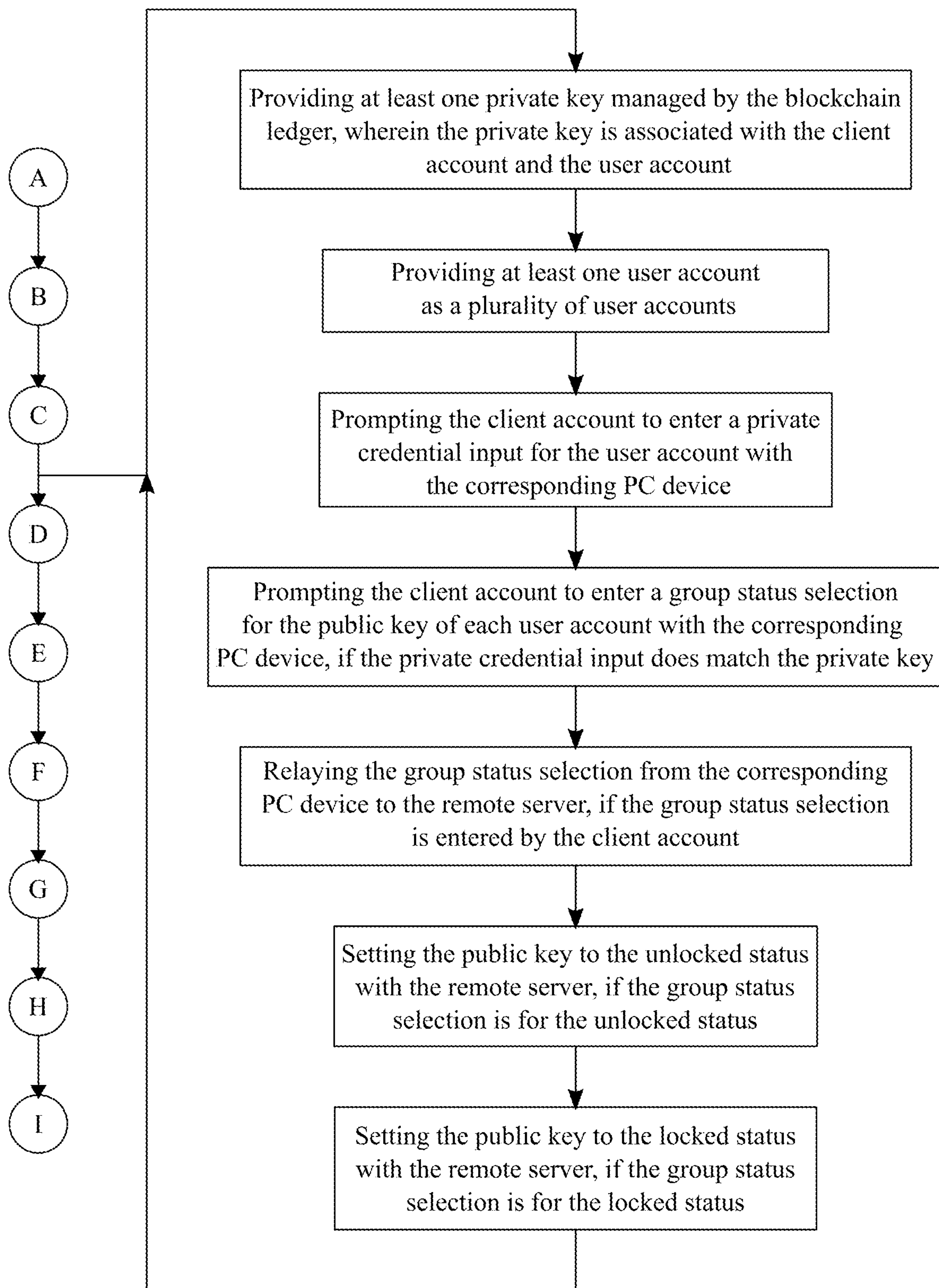


FIG. 5

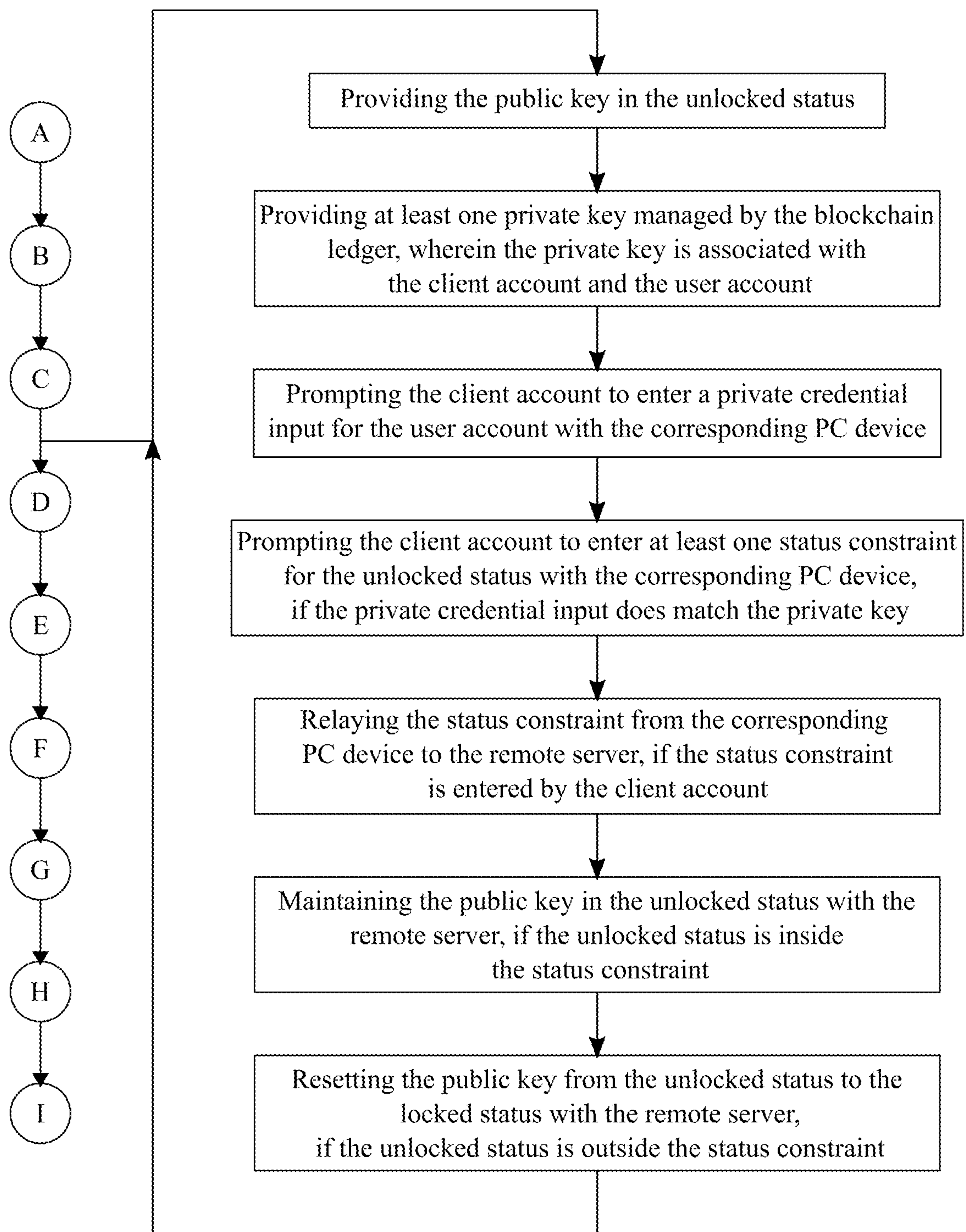


FIG. 6

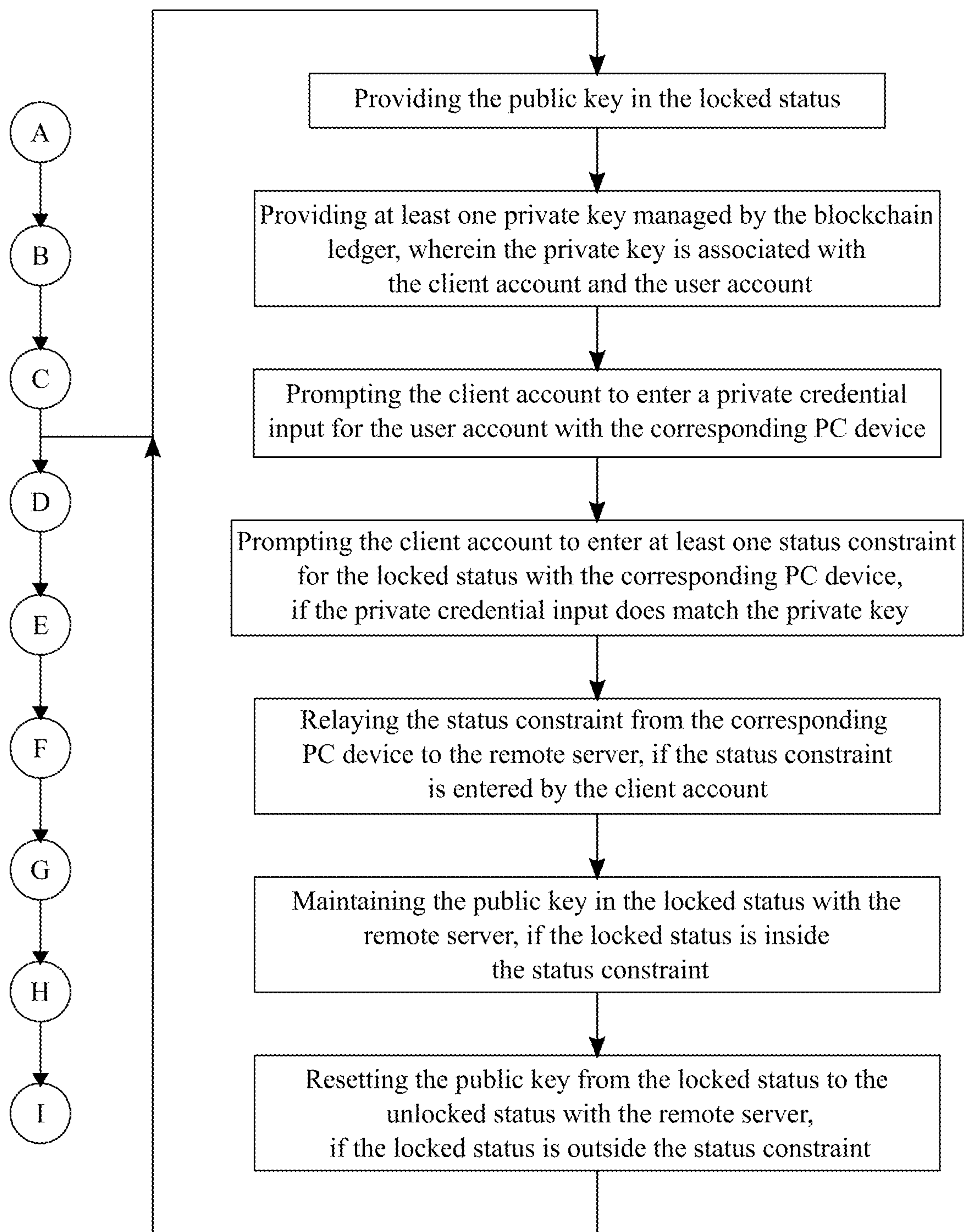


FIG. 7

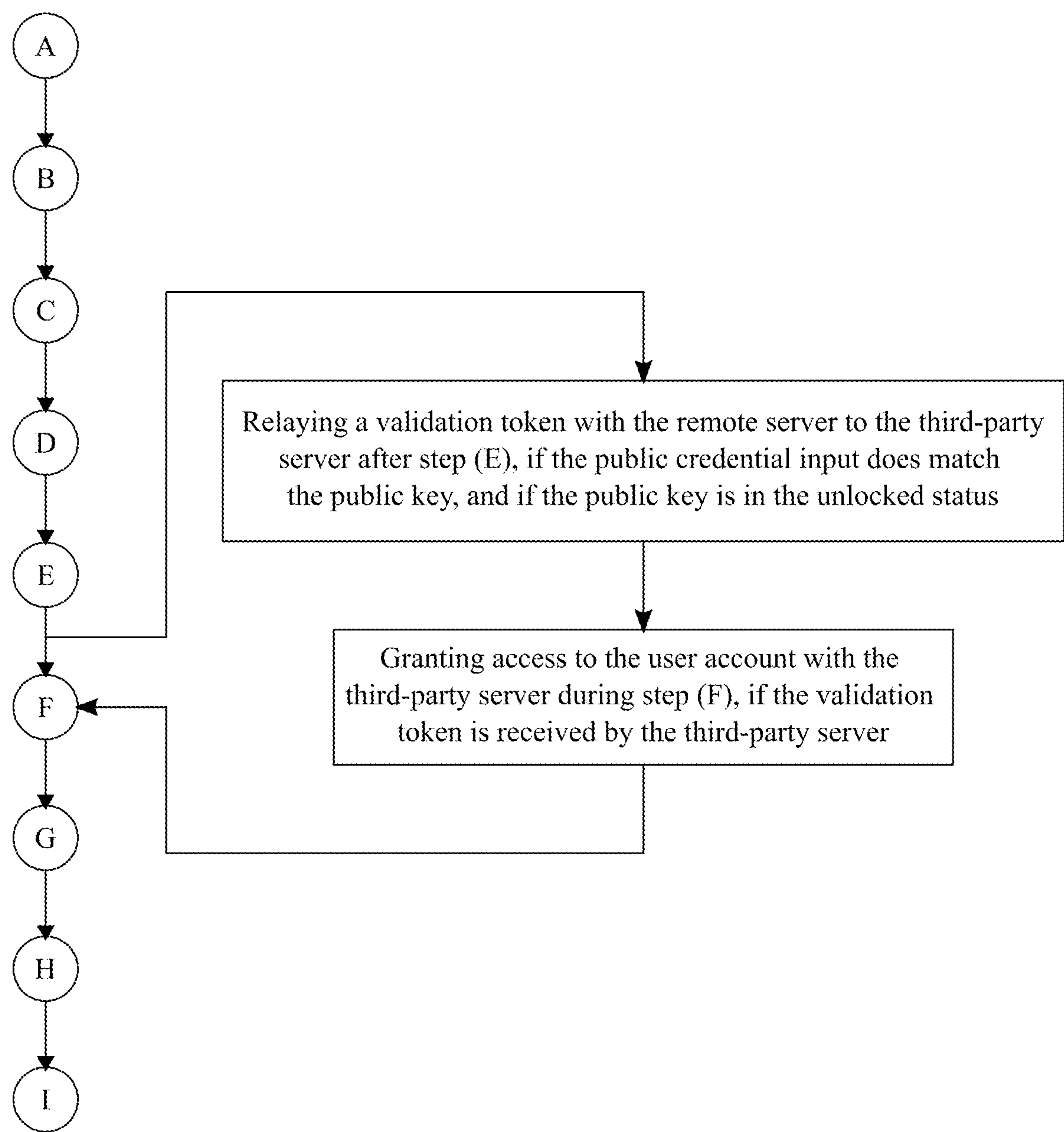


FIG. 8

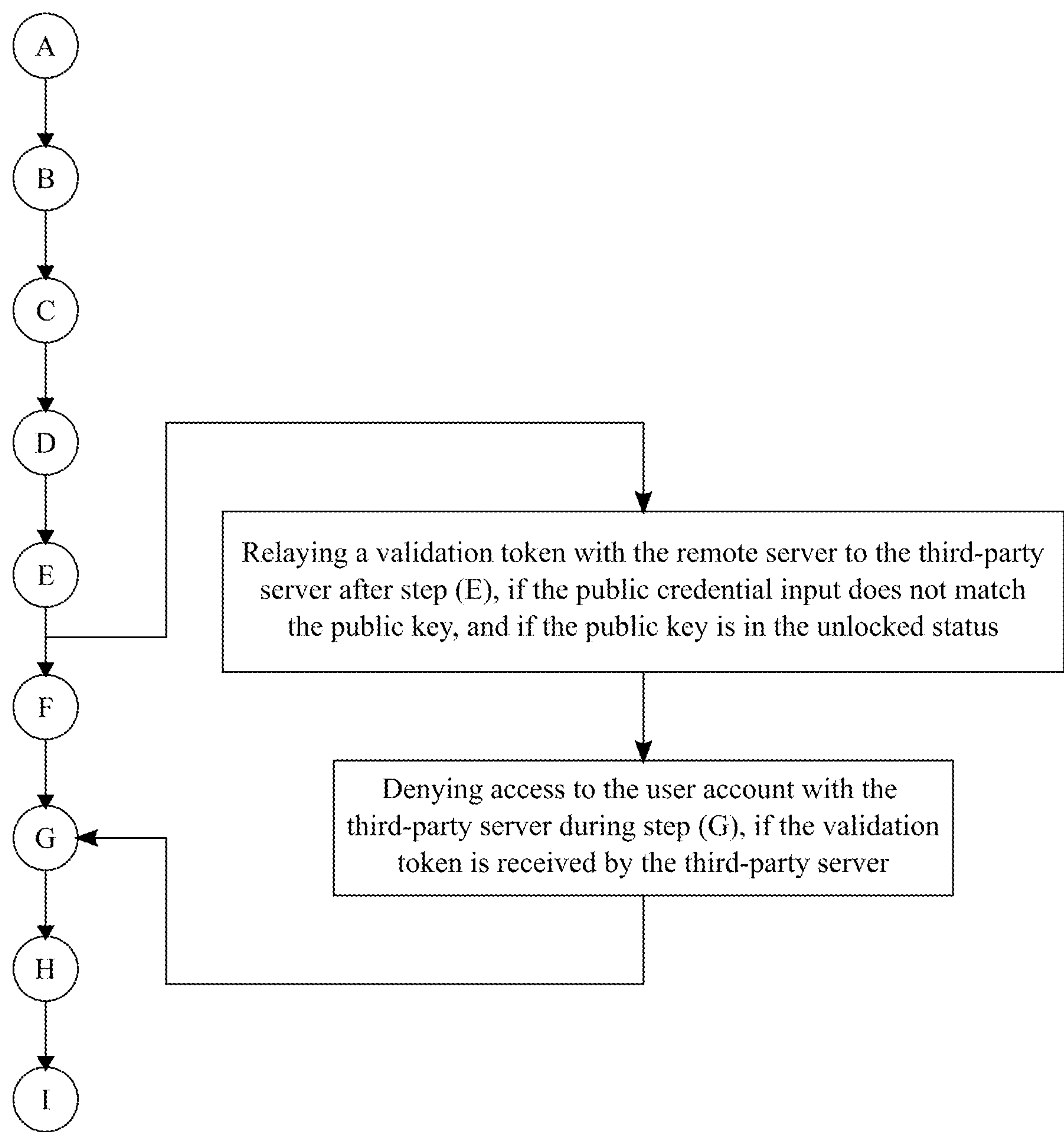


FIG. 9

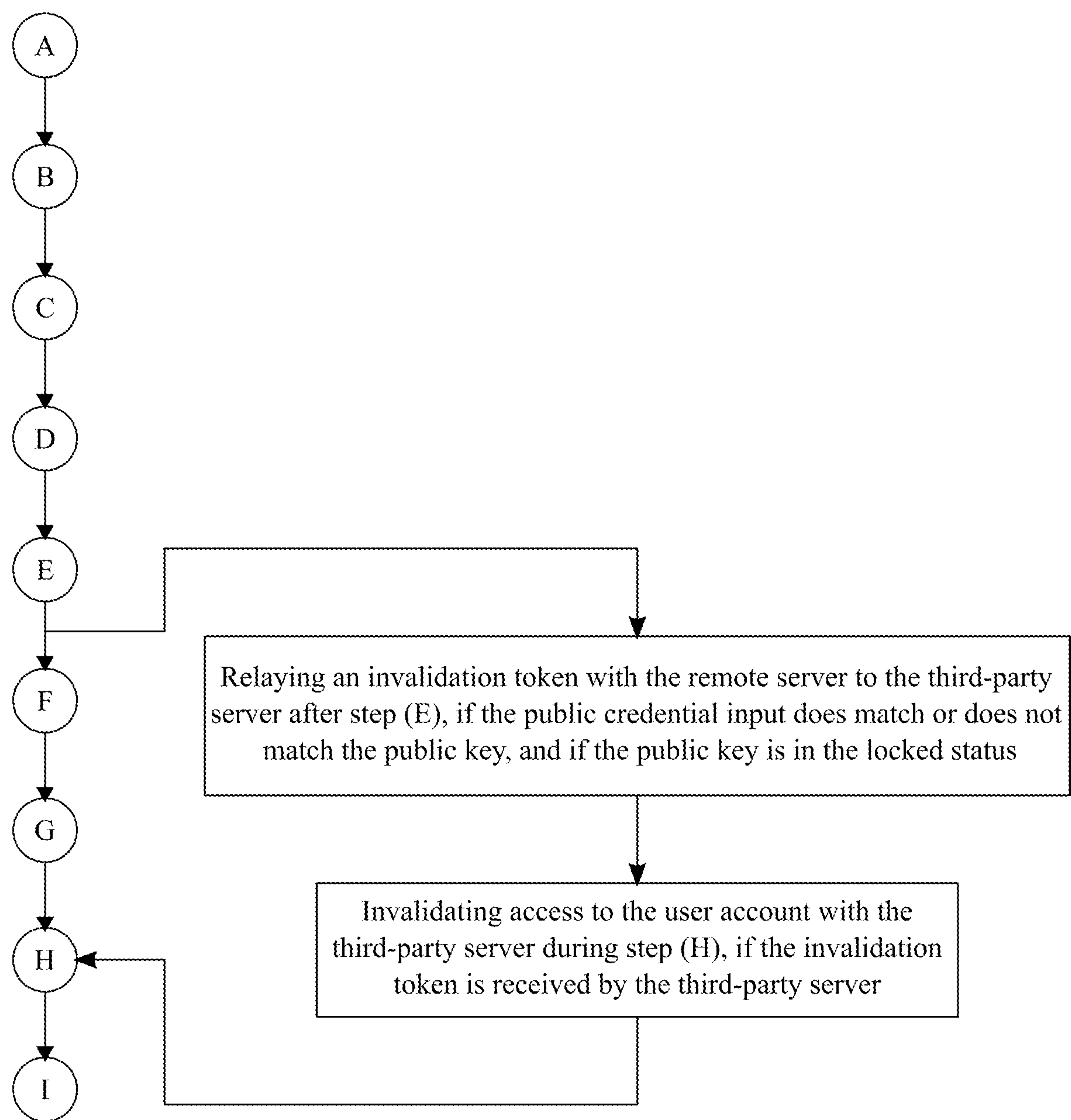


FIG. 10

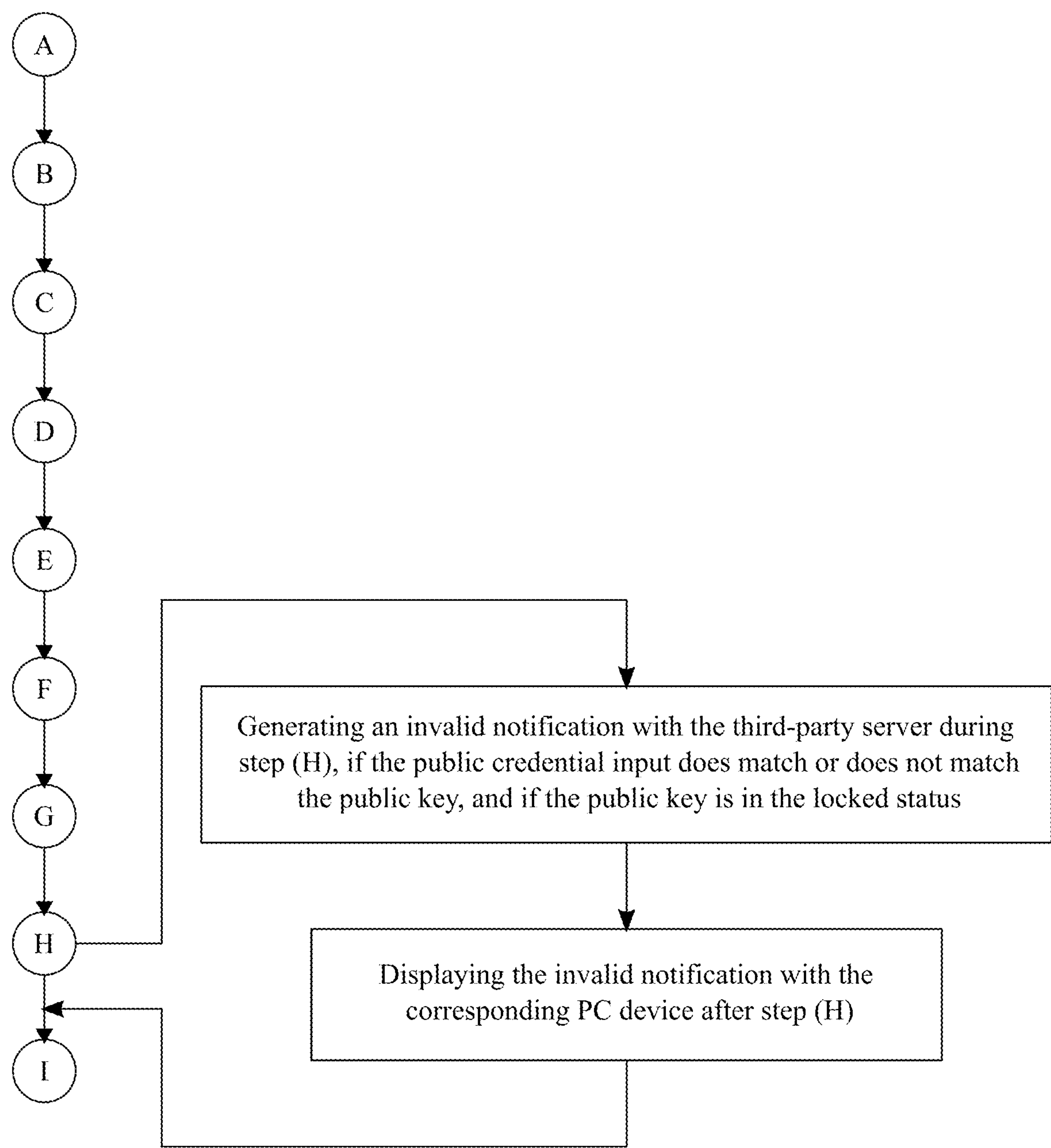


FIG. 11

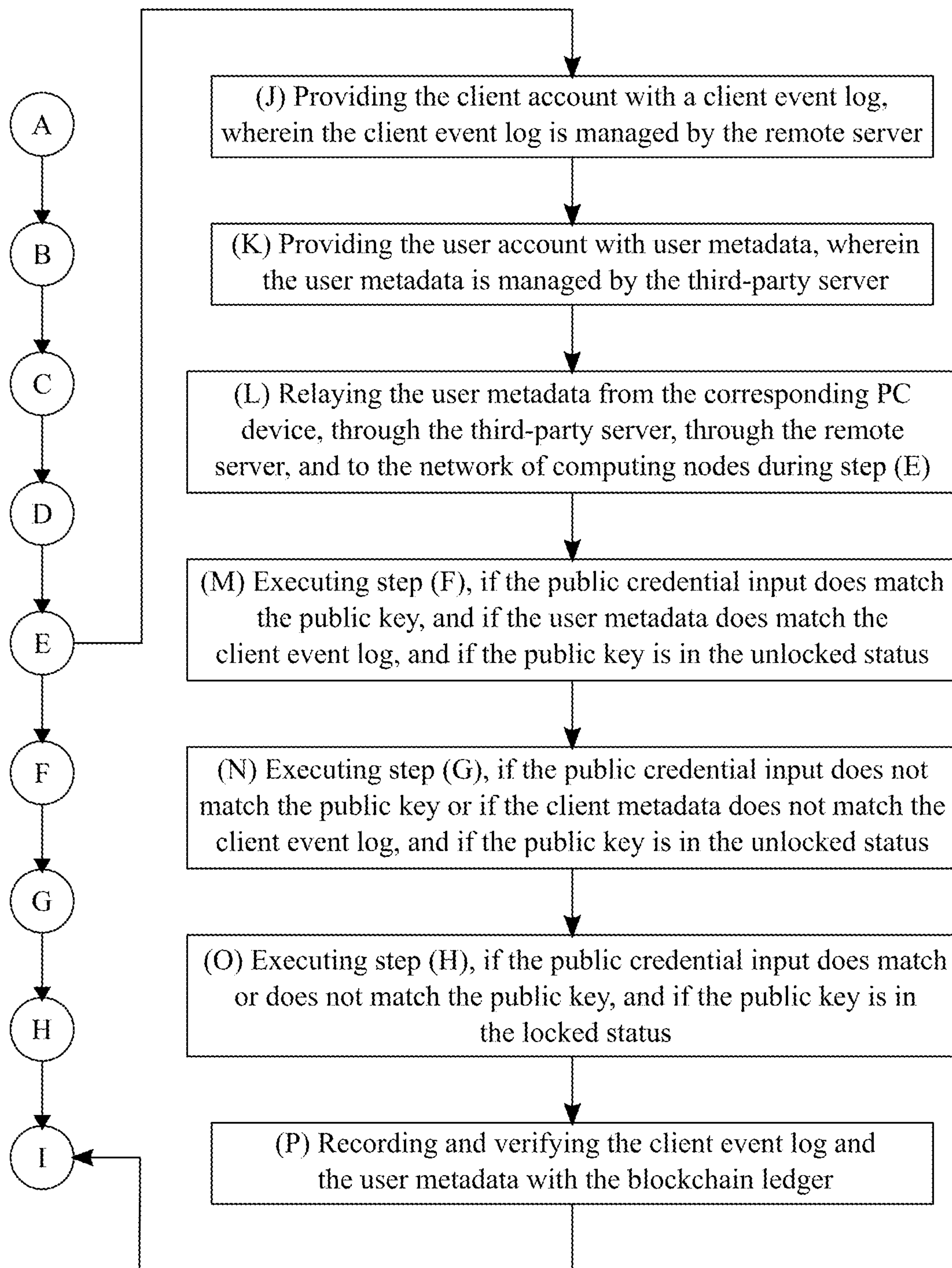


FIG. 12

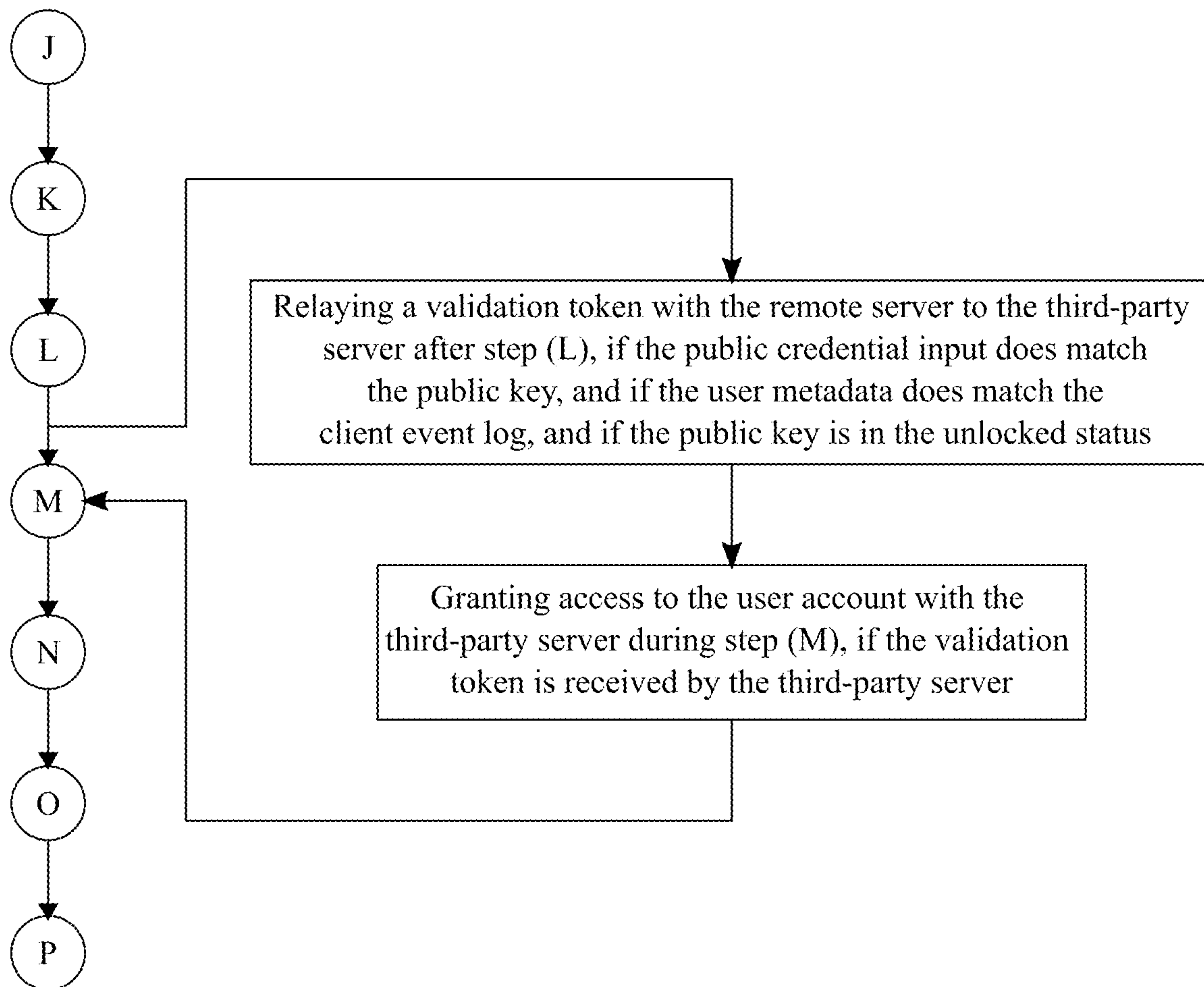


FIG. 13

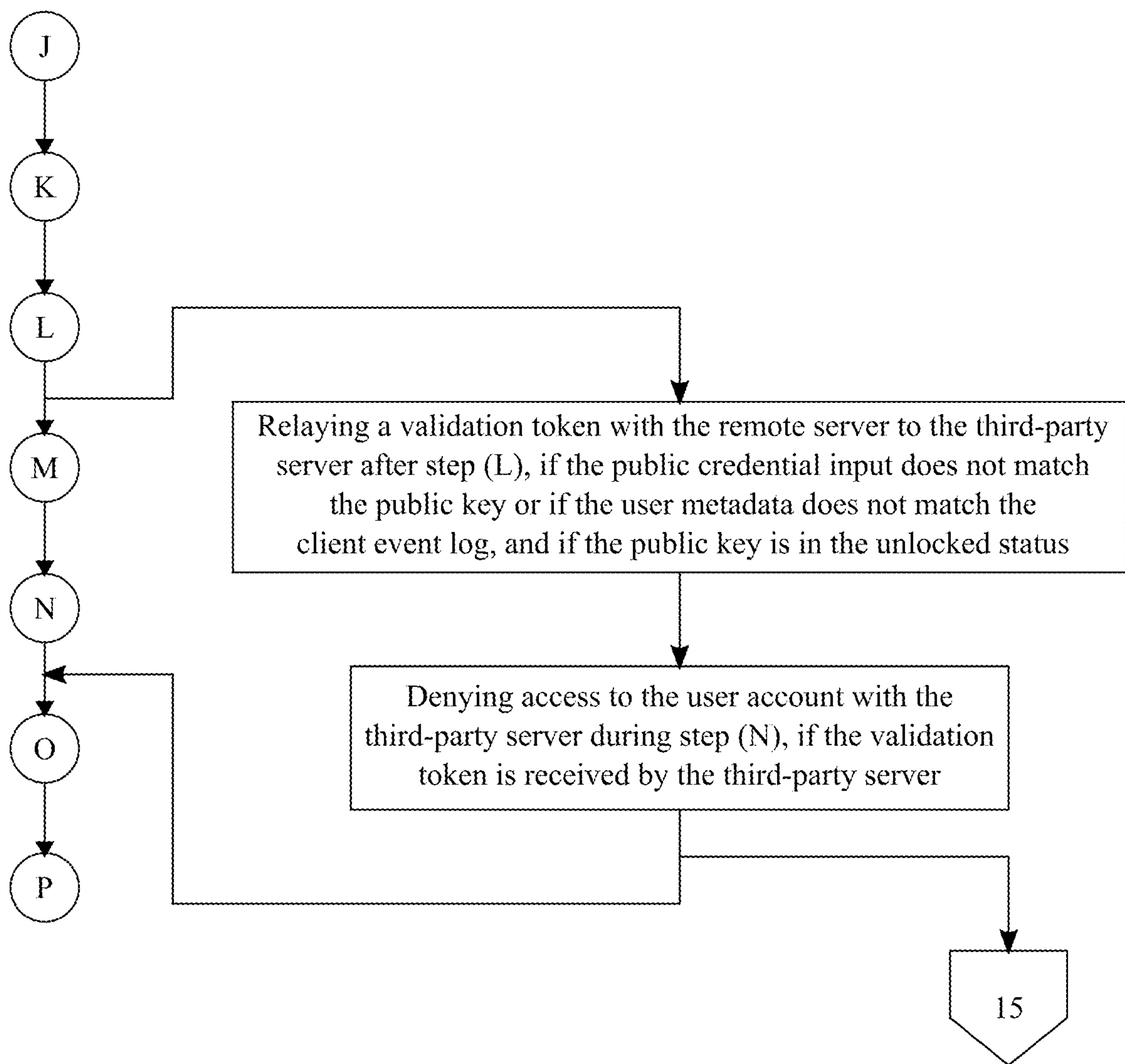


FIG. 14

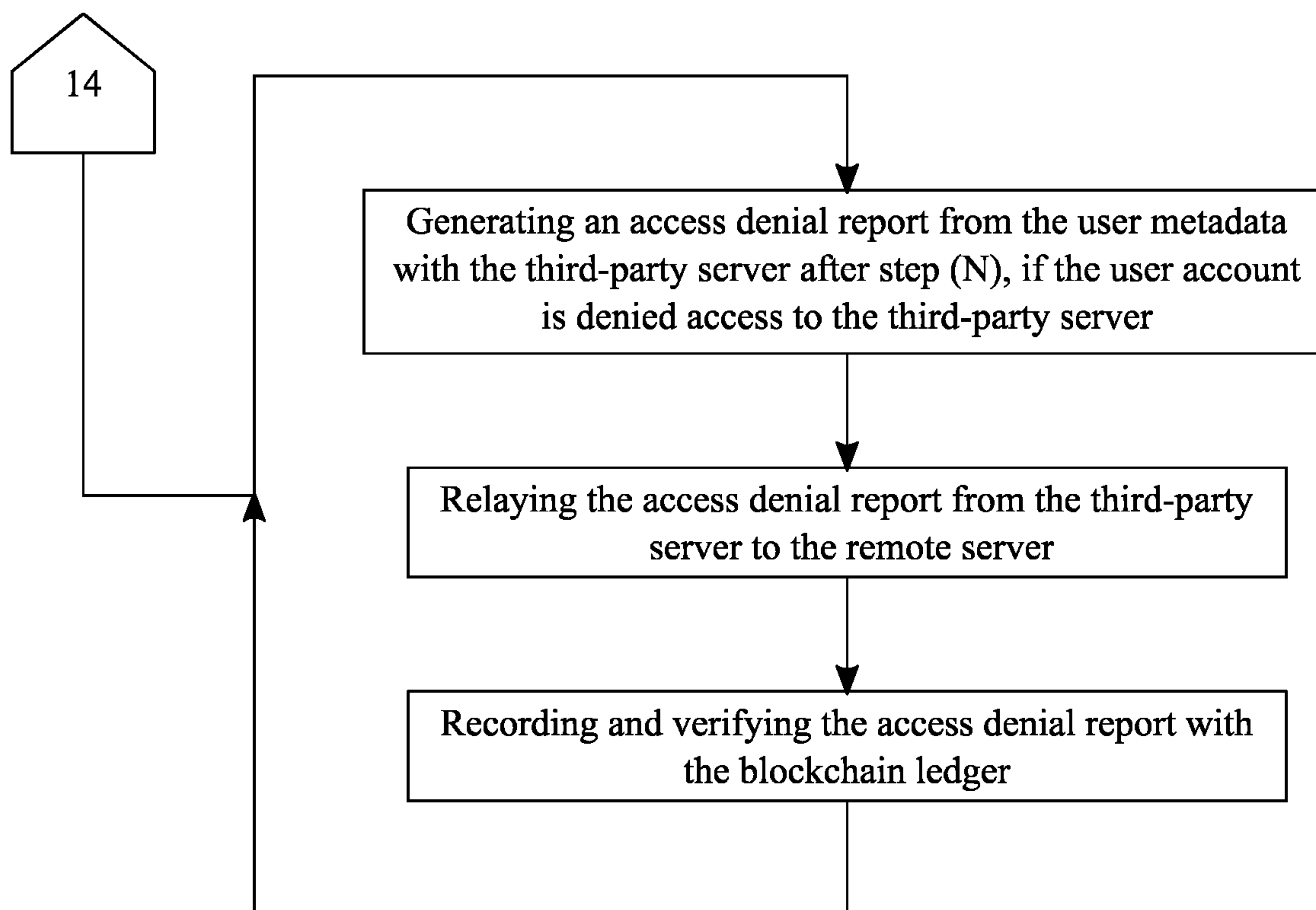


FIG. 15

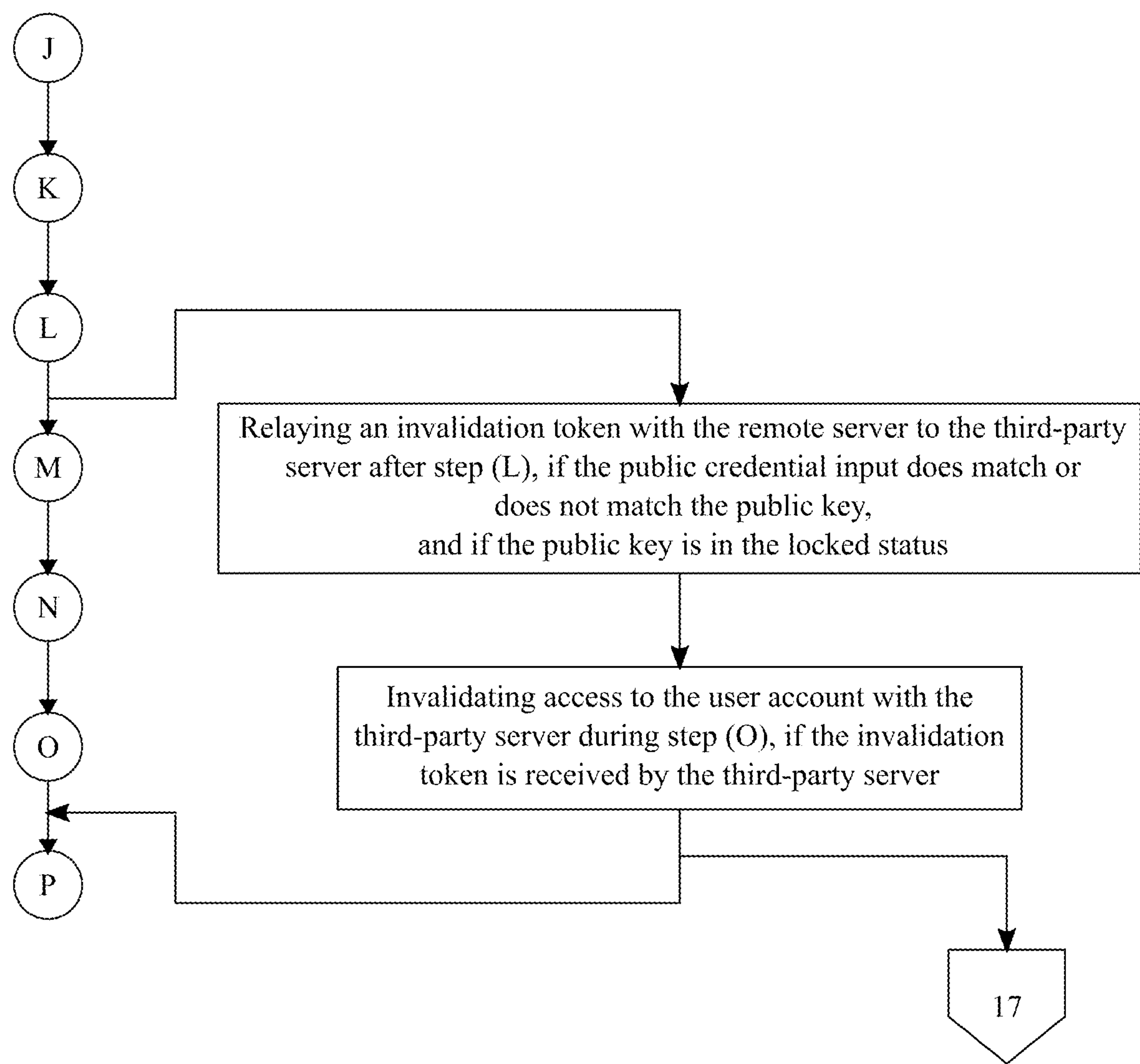


FIG. 16

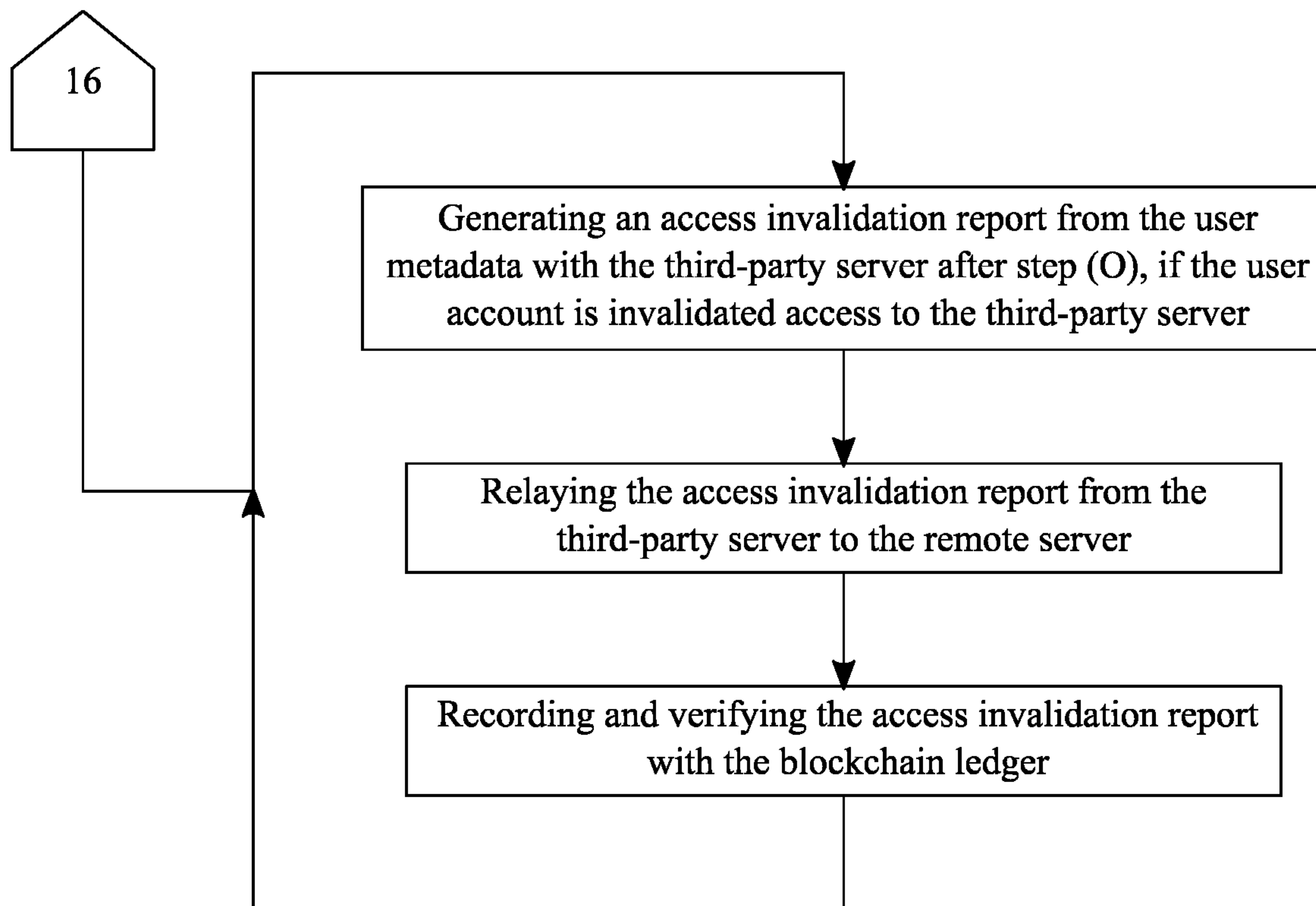


FIG. 17

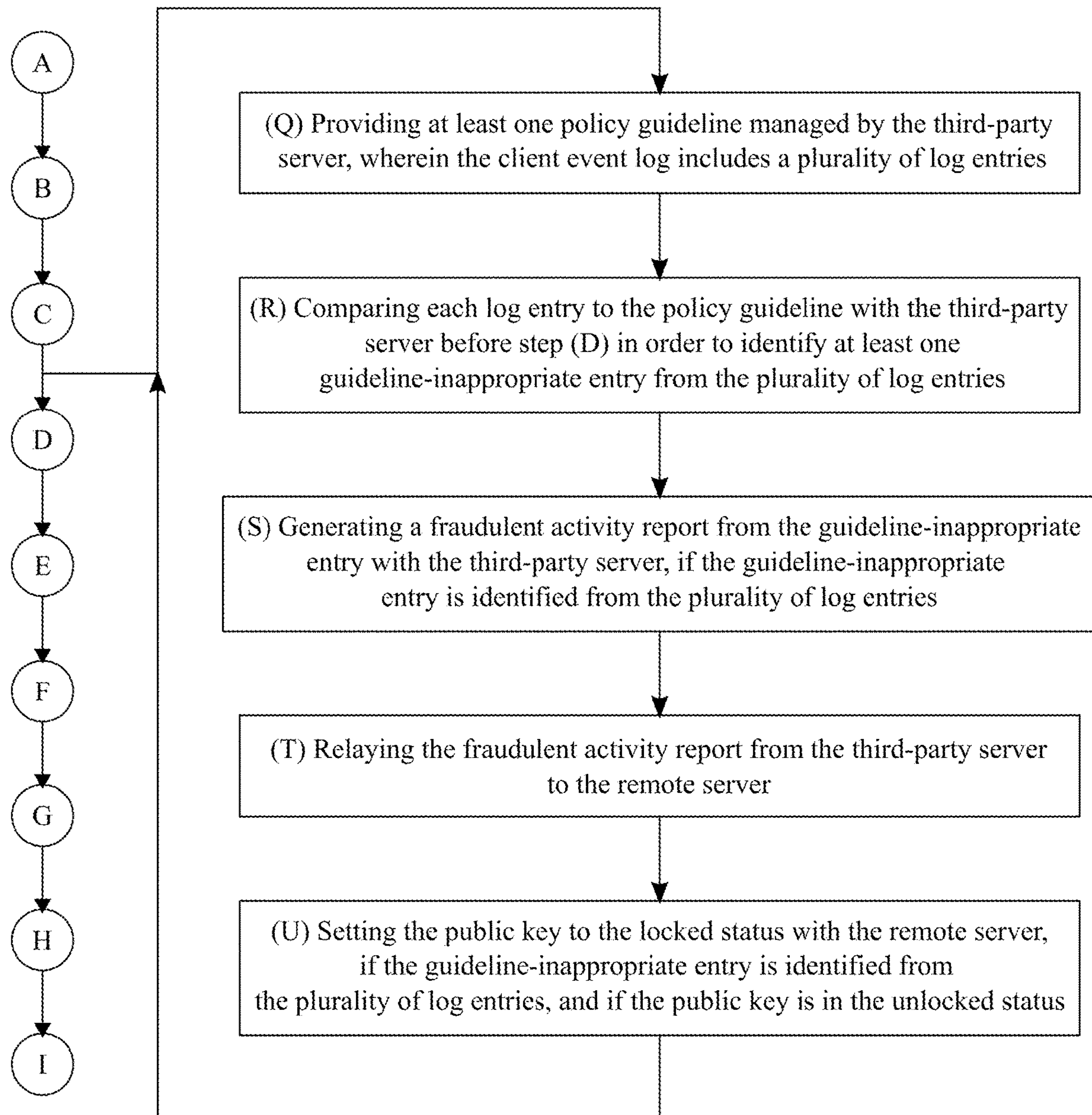


FIG. 18

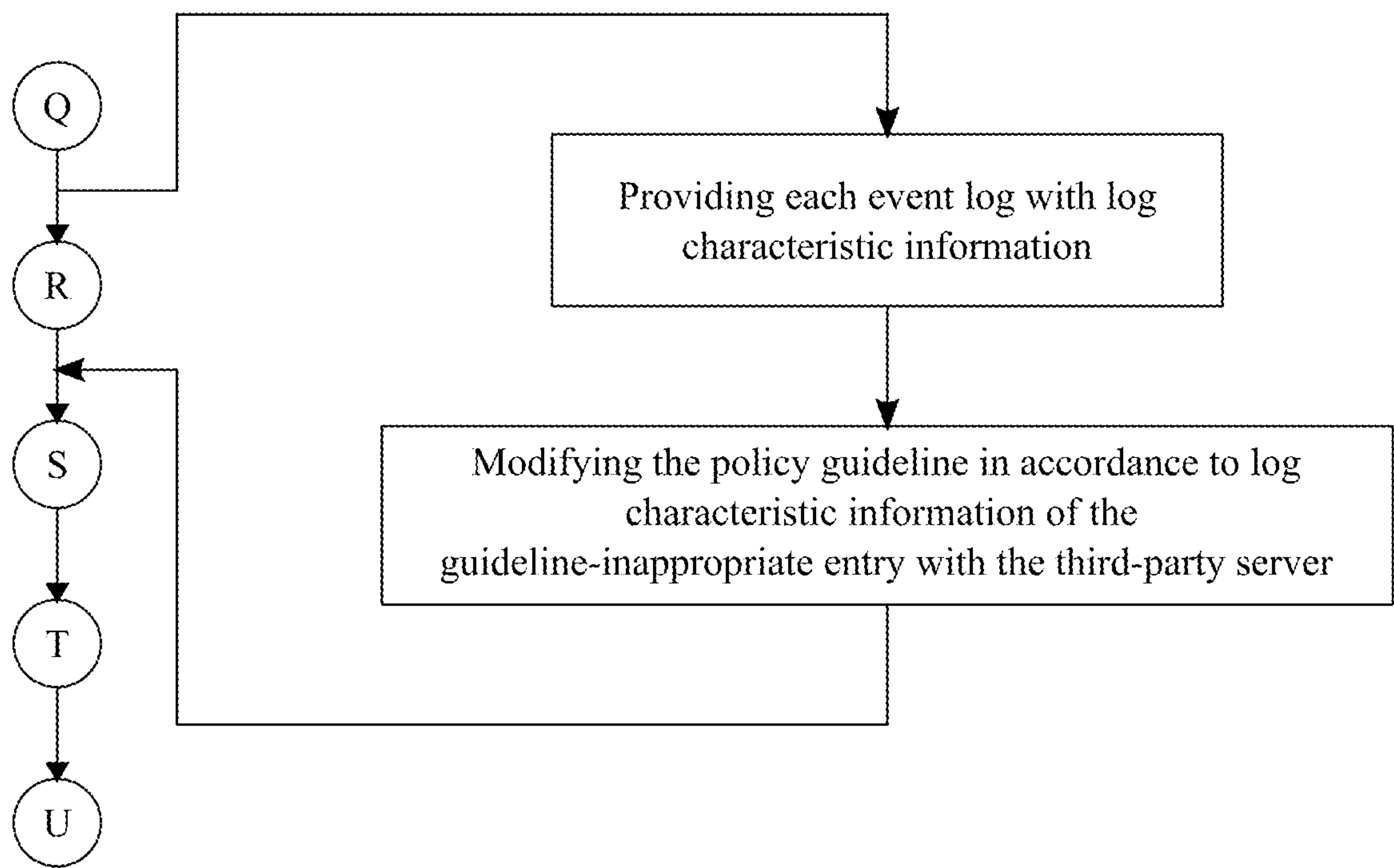


FIG. 19

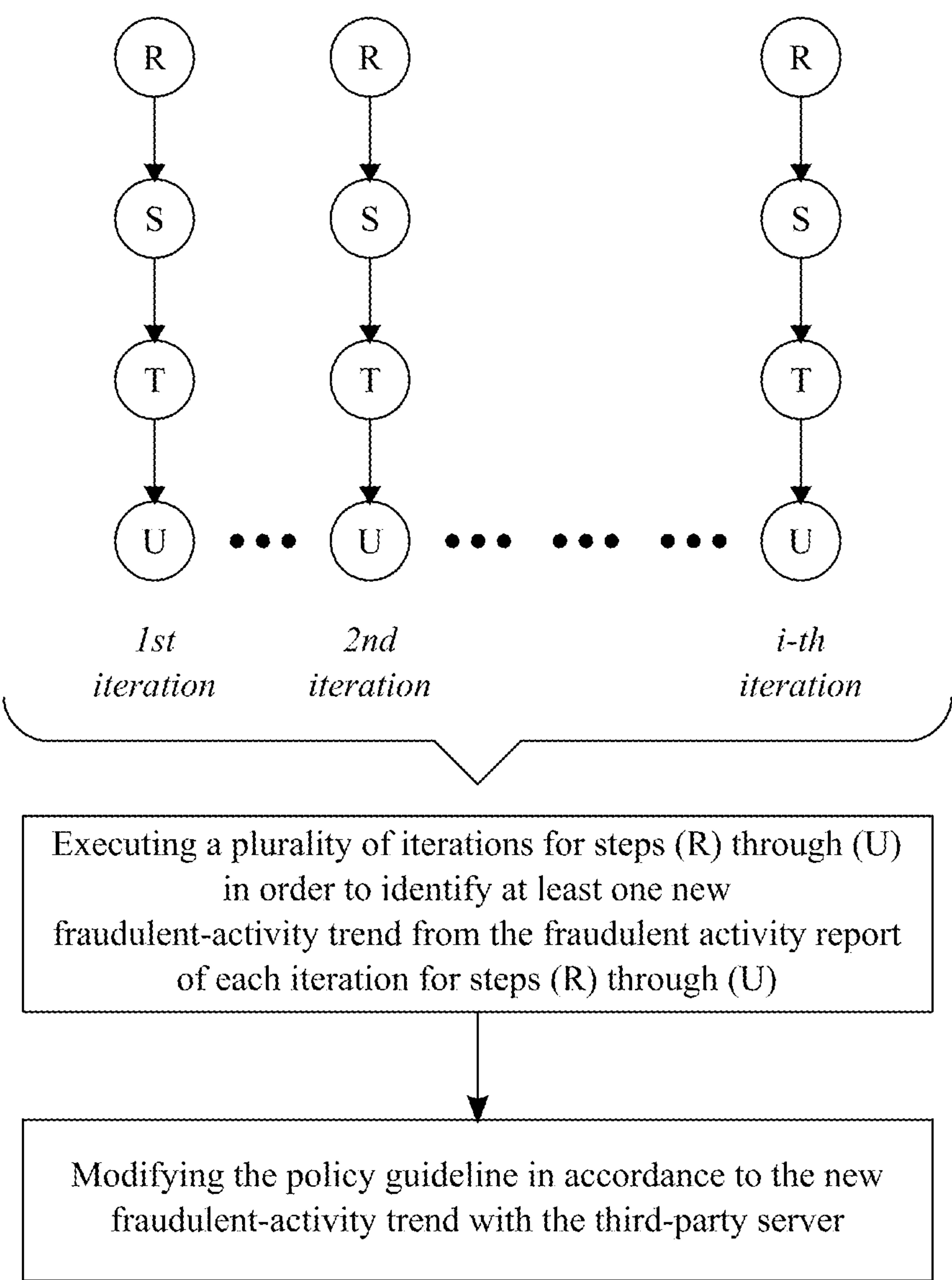


FIG. 20

SYSTEM AND METHOD TO FACILITATE AN ACCOUNT PROTECTION CHECK THROUGH BLOCKCHAIN

The current application is a continuation-in-part (CIP) application of a U.S. non-provisional application Ser. No. 17/710,738 filed on Mar. 31, 2022. The U.S. non-provisional application Ser. No. 17/710,738 claims a priority to a U.S. non-provisional application Ser. No. 17/207,411 filed on Mar. 19, 2021. The U.S. non-provisional application Ser. No. 17/207,411 claims a priority to a U.S. provisional application Ser. No. 62/991,990 filed on Mar. 19, 2020.

FIELD OF THE INVENTION

The present invention relates generally to data processing and account security. More specifically, the present invention is a system and a method to facilitate the validation of sets of credentials to grant access to secure accounts by utilizing blockchain technology.

BACKGROUND OF THE INVENTION

Username/Password credential validation methods are the most commonly used authentication methods on the Internet. Almost all online services, sites, applications, and processes rely on the validation of credentials prior to authorizing access to resources. The services, applications, and other interfaces rely on authentication prior to granting access; therefore, these systems keep tracking authentication requests continuously. This often leaves these systems open to receive requests for credential validation outside of predetermined parameters the actual owner of the credentials original intended. With the rise of identity theft, it is more common for people to have their credentials stolen and their personal accounts hacked. Therefore, there is a need for improved systems and methods to facilitate the validation of sets of credentials that may overcome one or more of the above-mentioned problems and/or limitations.

The present invention provides a system and method for facilitating the validation of sets of credentials by utilizing blockchain technology. The present invention allows a user to lock or unlock sets of credentials to further control access to one or more accounts using a blockchain ledger to prevent any other individual from accessing the accounts. The set of credentials is preferably provided as a public key and a private key that are recorded by the blockchain ledger and authenticated by a network of computing nodes. Before a service provider goes through an authentication process, the service provider requests a token from the service provided by the present invention. In addition, the service provider performs an automatic account protection check which analyses the validity of the request to ensure that the set of credentials has not been compromised. Based on if the credential set is in a locked or unlocked status, the present invention relays an invalidation token or a validation token to the service provider, respectively. If the invalidation token is relayed to the service provider, the service provider does not go through the authentication process. If the validation token is relayed to the service provider, the service provider proceeds with the authentication process. Additional benefits and features of the present invention are further discussed in the sections below.

SUMMARY OF THE INVENTION

This summary is provided to introduce a selection of concepts in a simplified form, that are further described

below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter. Nor is this summary intended to be used to limit the claimed subject matter's scope.

According to some embodiments, a method for facilitating an account protection check through blockchain is disclosed. Accordingly, the method may include a step of receiving, using a communication device, a login request associated with at least one user from a resource owner. The method may include a step of analyzing, using a processing device, the login request, which includes a public key recorded in a blockchain ledger. The login request may be analyzed utilizing user metadata to validate the authenticity of the login request. Further, the method may include a step of retrieving, using a storage device, a preauthorized token from the resource owner. In addition, the method may include a step of validating, using the processing device, the public key set state associated with the at least one user. The method may also include a step of generating, using the processing device, a token response associated with the credential set state. Further, the method may include a step of transmitting, using the communication device, the token response to the resource owner.

By leveraging the use of blockchain technology, the user is provided with a unique set of public key and private key, with the public key being tied to a credential set. The private key is required for the user to make any modifications to the data stored in the credential set. Using the private key, users can modify the credential set lock status. Any modification to the status of a credential set is published on the blockchain and validated by the network of computing nodes. The validation can happen via proof of stake or proof of share. The system can also leverage cryptocurrency and do a smart contract to leverage the upcoming proof of share from the cryptocurrency. Authentication providers can perform a lookup on the blockchain ledger to get the status of the credential set. Another advantage of the present invention is that it is nearly impossible to modify someone's credential set status, making the present invention highly secure form of account locking. A disadvantage of the present invention can be the time to update the credential lock statuses given the validation time required in the network of computing nodes. To speed this up, the present invention can utilize the fastest validation algorithms currently available.

Both the foregoing summary and the following detailed description provide examples and are explanatory only. Accordingly, the foregoing summary and the following detailed description should not be considered to be restrictive. Further, features or variations may be provided in addition to those set forth herein. For example, embodiments may be directed to various feature combinations and sub-combinations described in the detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram for the system of the present invention.

FIG. 2A is a flowchart illustrating the overall process for the method of the present invention.

FIG. 2B is a continuation of the flowchart illustrated in FIG. 2A.

FIG. 3 is a flowchart illustrating the subprocess of linking a user account to the client account.

FIG. 4 is a flowchart illustrating the subprocess of toggling between the unlocked status or the locked status for the credential set.

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FIG. 5 is a flowchart illustrating the subprocess of toggling between the unlocked status or the locked status for at least two or all of the credential sets for a plurality of user accounts.

FIG. 6 is a flowchart illustrating the subprocess of applying a status constraint for the unlocked status of the credential set.

FIG. 7 is a flowchart illustrating the subprocess of applying a status constraint for the locked status of the credential set.

FIG. 8 is a flowchart illustrating the subprocess of the remote server validating the credential set, because the credential set is in the unlocked status, in order for the third-party server to grant access to the user account.

FIG. 9 is a flowchart illustrating the subprocess of the remote server validating the credential set, because the credential set is in the unlocked status, in order for the third-party server to deny access to the user account.

FIG. 10 is a flowchart illustrating the subprocess of the remote server invalidating the credential set, because the credential set is in the locked status, in order for the third-party server to invalidate access to the user account.

FIG. 11 is a flowchart illustrating the subprocess of displaying that the user account is invalid.

FIG. 12 is a flowchart illustrating the subprocess of collecting the user metadata to further validate the access request to the user account.

FIG. 13 is a flowchart illustrating the subprocess of the remote server validating the credential set using the user metadata, because the credential set is in the unlocked status, in order for the third-party server to grant access to the user account.

FIG. 14 is a flowchart illustrating the subprocess of the remote server validating the credential set using the user metadata, because the credential set is in the unlocked status, in order for the third-party server to deny access to the user account.

FIG. 15 is a flowchart illustrating the subprocess of the generating an access denial report if the third-party server denies access to the user account.

FIG. 16 is a flowchart illustrating the subprocess of the remote server invalidating the credential set using the user metadata, because the credential set is in the locked status, in order for the third-party server to invalidate access to the user account.

FIG. 17 is a flowchart illustrating the subprocess of the generating an access invalidation report if the third-party server invalidates access to the user account.

FIG. 18 is a flowchart illustrating the subprocess of modifying the policy guideline from automatic detections of guideline-inappropriate entries to automatically set the credential set to the locked status.

FIG. 19 is a flowchart illustrating the subprocess of modifying the policy guideline from log characteristic information of the guideline-inappropriate entry.

FIG. 20 is a flowchart illustrating the subprocess of modifying the policy guideline from fraudulent-activity trends identified from the fraudulent activity report.

DETAIL DESCRIPTIONS OF THE INVENTION

All illustrations of the drawings are for the purpose of describing selected versions of the present invention and are not intended to limit the scope of the present invention.

The present invention provides a system and method for facilitating an account protection check through blockchain. More specifically, the present invention utilizes blockchain

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technology to record and authenticate the access request from a user. Further, the present invention automatically performs an account protection check when the user desires to access one or more accounts based on the activities of the user during the login request. The account protection check provides greater security to the accounts in addition to the set of credentials lock status. For example, if the login request is performed from a foreign location or a computing device not usually utilized by the user, the login request is automatically blocked to prevent an unwanted party from hacking the certain user account. In addition, by utilizing blockchain technology, only the user can have access to the one or more accounts by providing the set of credentials as a public key and a private key generated by using the blockchain technology. With reference to FIG. 1, the system of the present invention includes at least one remote server that manages at least one client account (Step A). The client account is associated with a corresponding personal computing (PC) device, which can be, but is not limited to, a desktop computer, a notebook computer, a smartphone, or a mobile tablet. The remote server is a computing node where information provided by the client account is processed and stored.

Additionally, with reference to FIG. 1, the system of the present invention includes at least one third-party server that manages at least one user account (Step B). The user account is associated with the corresponding PC device that the user utilizes to access the user account. The third-party server is a computing node that processes and stores information provided by the user account. Moreover, the third-party server is a computing node that is associated with a third-party service, which can be, but is not limited to, an email service, an electronic vehicle entry service, a building access service, a computer operating system, or an Internet of Things (IoT) network. Further, the user account is associated with the client account. The user account and the client account are linked so that the user can access the resources associated with the client account. Furthermore, the system of the present invention includes at least one blockchain ledger that manages at least one public key (Step C). The blockchain ledger is a distributed database that stores information related to the public key and the client account. The public key is a cryptographic key that can be used by the user to perform transactions recorded on the blockchain ledger. Accordingly, the public key is associated with the client account and the user account, and the public key can be in either an unlocked status or a locked status. The unlocked status or the locked status can be toggled by the user. In the unlocked status, the user is allowed to access the client account, while in the locked status the user is not allowed to access the client account. In addition, the blockchain ledger is managed by a network of computing nodes to decentralize the verification process of the public key, virtually preventing any unauthorized access into the client account.

With reference to FIG. 2, the method of the present invention follows an overall process for facilitating the use of blockchain technology to protect one or more accounts. The overall process begins by prompting the client account to enter a public credential input for the user account with the corresponding PC device (Step D). More specifically, the client account is prompted to enter the public key in order to access the service provided by the third-party server. The public credential input is relayed from the corresponding PC device, through the third-party server, through the remote server, and to the network of computing nodes, if the public credential input is entered by the client account (Step E). The public credential input is received by the third-party server

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for an authentication process. However, before the public credential input is authenticated, the public credential input is sent to the remote server in order to check if the credential set is in the unlocked or locked status. The third-party server grants access to the user account, if the public credential input does match the public key, and if the public key is in the unlocked status (Step F). In further detail, the network of computing nodes validates the public credential input because the public key is in the unlocked status. Therefore, the third-party server can proceed with the authentication process. Further, the third-party server authenticates the public credential input because the public credential input matches the public key, and, thus, the user can access the service provided by the third-party account.

Alternatively, in reference to FIG. 2, the third-party server denies access to the user account, if the public credential input does not match the public key, and if the credential set is in the unlocked status (Step G). Similar to Step F, the remote server validates the public credential set because the public key is in the unlocked status in order for the third-party server to proceed with the authentication process. However, the third-party server fails to authenticate the public credential input because the public credential input does not match the public key. For example, the public credential input may have included the wrong numbers. The third-party server follows standard authentication procedure by prompting to reenter the public credential input. As another alternative result, the third-party server invalidates access to the user account, if the public credential input does or does not match the public key, and if the public key is in the locked status (Step H). In further detail, the remote server invalidates the public credential input because the public key is in the locked status. This prevents the third-party server from proceeding with the authentication process, even if the public credential input is correct. Instead of going through the authentication process, the third-party server invalidates access to the user account by providing an error message or error website page to the user. Thus, an unwanted user cannot attempt to hack the user account when the credential set is in the locked status. For further security, the overall process concludes by recording and verifying Steps F, G, and H are with the blockchain ledger (Step I). This provides greater security as the information recorded in the blockchain ledger is unalterable.

The system of the present invention provides at least one private key managed by the blockchain ledger. The private key is associated with the client account and the user account. Unlike the public key, the private key preferably enables the user to make changes to the client account as well as to perform other administrative tasks, such as claiming ownership to the blockchain ledger or linking the user account to the client account. In order for the user to link the user account with the client account and with reference to FIG. 3, the following subprocess is executed. The corresponding PC device prompts the client account to enter a private credential input for validation purposes. Then, the corresponding PC device prompts the client account to link to the user account during the Step B, if the private credential input does match the private key. In more detail, the user is prompted to link one or more user accounts from various third-party services. This can be executed in various conditions. For example, the user can be prompted to link an existing user account for a service to the client account or the user can be prompted to link a new user account for a service to the client account during the account creation process hosted by the third-party server. The remote server generates an association between the user account and

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the client account, if the user account is selected to be linked to the client account. The association between the user account and the client account allows the remote server to validate or invalidate the credential set depending on if the credential set is in the unlocked status or the locked status. Further, the association between the user account and the client account provides the user the ability to unlock or lock the credential set.

In order for the user to toggle between the unlocked or locked status for the public key and with reference to FIG. 4, the following subprocess is executed. The corresponding PC device prompts the client account to enter a private credential input for validation purposes as the toggling between the unlocked or locked status for the public key is an administrative task. Then, the corresponding PC device prompts the client account to enter a single status selection for the public key, if the private credential input does match the private key. In further detail, the single status selection is an input to toggle the public key to either the unlocked status or the locked status. The single status selection is relayed from the corresponding PC device to the remote server, if the single status selection is entered by the client account. Thus, the remote server is instructed to toggle the status of the public key based on the single status selection. The remote server sets the credential set to the unlocked status, if the single status selection is for the unlocked status. Thus, the user can toggle the public key from the locked status to the unlocked status as desired to prevent unauthorized access. Alternatively, the remote server sets the public key to the locked status, if the single status selection is for the locked status. Thus, the user can toggle the public key from the unlocked status to the locked status.

In order for the user to toggle between the unlocked status or the locked status for at least two or all of the public keys for a plurality of user accounts and with reference to FIG. 5, the following subprocess is executed. The at least one user account is provided as a plurality of user accounts. The corresponding PC device prompts the client account to enter a private credential input for validation purposes as the toggling between the unlocked or locked status for the public keys is an administrative task. The corresponding PC device prompts the client account to enter a group status selection for the public key for each user account, if the private credential input does match the private key. In further detail, the group status selection is an input to toggle at least two or all of the public keys for the plurality of user accounts to either the unlocked status or the locked status. The group status selection is relayed from the corresponding PC device to the remote server, if the group status selection is entered by the client account. Thus, the remote server is instructed to toggle the status of at least two or all of the public keys based on the group status selection. The remote server sets the public key of each user account to the unlocked status, if the group status selection is for the unlocked status. Thus, the user can toggle at least two or all of the public keys for the plurality of user accounts from the locked status to the unlocked status. Alternatively, the remote server sets the public key of each user account to the locked status, if the group status selection is for the locked status. Thus, the user can toggle at least two or all of the public keys for the plurality of user accounts from the unlocked status to the locked status.

In order for the user to place a status constraint for the unlocked status of the credential set and with reference to FIG. 6, the following subprocess is executed. The corresponding PC device prompts the client account to enter a private credential input for validation purposes as placing a

status constraint for the unlocked status is an administrative task. With the credential set already set to the unlocked status, the corresponding PC device prompts the client account to enter at least one status constraint for the unlocked status, if the private credential input does match the private key. The status constraint can be, but is not limited to, a time-based constraint, a location-based constraint, a subscription-based constraint, or a combination thereof. For example, if the status constraint is a time-based constraint, the user can set the public key to be in the unlocked status for specified period of time. In another example, if the status constraint is a location-based constraint, the user can set the public key to be in the unlocked status whenever the user is in a specified location. In further detail, a geofenced area may be generated, and the public key would stay in the unlocked status or toggle to the unlocked status whenever the corresponding PC device is within or enters the generated geofenced area. This can be managed and monitored through a Global Navigation Satellite System (GNSS) device, which may be provided with the corresponding PC device. Moreover, if the status constraint is a subscription-based constraint, the user can set the public key to be in the unlocked status for as long as a subscription to a service is active. The status constraint is relayed from the corresponding PC device to the remote server, if the status constraint is entered by the client account. Thus, the remote server is prepared to set the status constraint for the unlocked status. The remote server maintains the public key in the unlocked status, if the unlocked status is inside the status constraint. In further detail of the aforementioned examples, the public key remains in the unlocked status, if a specified period of time is still ongoing, if the corresponding PC device is within a specified geofenced area, if the subscription to a service is active, or if a combination thereof. The remote server resets the public key from the unlocked status to the locked status, if the unlocked status is outside the status constraint. In further detail of the aforementioned examples, the public key is toggled from the unlocked status to the locked status, if the specified period of time has lapsed, if the corresponding PC device is outside a specified geofenced area, if the subscription to a service is canceled or expired, or if a combination thereof. Thus, the user is able to place a status constraint for the unlocked status of the public key.

Alternatively, and with reference to FIG. 7, the user can place a status constraint for the locked status of the credential set through the following subprocess. The corresponding PC device prompts the client account to enter a private credential input for validation purposes as placing a status constraint for the locked status is an administrative task. With the credential set already set to the locked status, the corresponding PC device prompts the client account to enter at least one status constraint for the locked status, if the private credential input does match the private key. The status constraint can be any type of constraint such as those aforementioned examples. The status constraint is relayed from the corresponding PC device to the remote server, if the status constraint is entered by the client account. Thus, the remote server is prepared to set the status constraint for the locked status. The remote server maintains the public key in the locked status, if the locked status is inside the status constraint. In further detail of the aforementioned examples, the public key remains in the locked status, if a specified period of time is still ongoing, if the corresponding PC device is within a specified geofenced area, if the subscription to a service is inactive, or if a combination thereof. The remote server resets the public key from the unlocked status

to the locked status, if the unlocked status is outside the status constraint. In further detail of the aforementioned examples, the public key is toggled from the locked status to the unlocked status, if the specified period of time has lapsed, if the corresponding PC device is outside a specified geofenced area, if the subscription to a service is reactivated, or if a combination thereof. Thus, user is able to place a status constraint for the locked status of the public key.

In order for the remote server to validate the public key before authentication is processed by the third-party server and with reference to FIG. 8, the following subprocess is executed. Before the third-party server can authenticate the public credential input provided by the client account, the third-party server requests a validation token from the remote server to check the status of the public key. The validation token can be a data package including the information regarding the current lock status of the public key. The validation token is relayed with the remote server to the third-party server after Step E, if the public credential input does match the public key, and if the public key is in the unlocked status. The third-party server can then proceed with the authentication process. The third-party server grants access to the user account during Step F, if the validation token is received by the third-party server. In further detail, access is granted because the public credential input matches the public key after the public credential input is validated by the network of computing nodes. Therefore, the public credential input passed the authentication process, and the public key is validated to be in the unlocked status.

Alternatively, and with reference to FIG. 9, the validation token is also relayed with the remote server to the third-party server after Step E, if the public credential input does not match the public key, and if the public key is in the unlocked status. In this situation, since the public key is in the unlocked status, the third-party server can proceed with the authentication process. However, the third-party server denies access to the user account during Step G, if the validation token is received by the third-party server, due to the public credential input not matching the public key. In further detail, access is denied because the public credential input does not match the public key, and therefore, the public credential input failed the authentication preprocess by the network of computing nodes, and the public key was validated to be in the unlocked status.

In order for the remote server to invalidate the credential set before authentication is processed by the third-party server, and with reference to FIG. 10, the following subprocess is executed. Before the third-party server can authenticate the public credential input provided by the client account, the third-party server requests a token from the remote server. An invalidation token is relayed with the remote server to the third-party server after Step E, if the public credential input does match or does not match the credential set, and if the public key is in the locked status. Unlike the validation token, the invalidation token can be a data package including information regarding the lock status of the public key when the public key has been set to locked status. Since the public key is in the locked status, the third-party server does not proceed with authentication process. The third-party server invalidates access to the user account during Step H if the invalidation token is received by the third-party server. In further detail, the user account is prevented from being accessed altogether because the public key is in the locked status, which prevents access to the client account by any user even if the public credential input matches the public key.

Moreover, and with reference to FIG. 11, the third-party server generates an invalid notification during Step H, if the public credential input does match or does not match the public key, and if the public key is in the locked status. The invalid notification can be, but is not limited to, an error message or an error website page which prevents access to the user account. The invalid notification is then displayed with the corresponding PC device after Step H. In further detail, the invalid notification can be displayed as a pop-up error message after the user is prevented access to the user account or the user is directed to an error website page.

With reference to FIG. 12, the system of the present invention includes at least one remote server that manages a client event log (Step J). The client event log is a compilation of metadata that is used to form a profile of the user to further validate the login request when the user is trying to access the client account. Additionally, the user account is provided with user metadata that is managed by the third-party server (Step K). The user metadata includes, but is not limited to, data regarding the corresponding PC device, the usual geographic location of the user, and the usual timing when the user access the client account. By using the client event log for each account protection check, several patterns, rules, and machine learning algorithms can be leveraged to automatically recognize, isolate, block, and/or alert on ingress calls from potential fraudulent actors. To do so, the user metadata is relayed from the corresponding PC device, through the third-party server, through the remote server, and to the network of computing nodes during Step E to prevent alterations to the history of the client event log (Step L). The collected user metadata can then be leveraged to allow or prevent access to the user account. If the public credential input does match the public key, and if the user metadata does match the client event log, and if the public key is in the unlocked status, Step F is executed, which grants access to the user account by the third-party server (Step M). If the public credential input does not match the public key or if the client metadata does not match the client event log, and if the public key is in the unlocked status, Step G is executed, which denies access to the user account by the third-party server (Step N). Further, if the public credential input does match or does not match the public key, and if the public key is in the locked status, Step H is executed, which invalidates access to the user account by the third-party server (Step O). Throughout this subprocess, the client event log and the user metadata are recorded and verified with the blockchain ledger for greater security scrutiny (Step P).

With reference to FIG. 13, the user metadata can be utilized for greater identity verification in addition to the use of the public key. Before the third-party server can authenticate the public credential input provided by the client account, the third-party server requests a validation token from the remote server to check the status of the public key. The validation token is relayed with the remote server to the third-party server after Step L, if the public credential input does match the public key, if the user metadata does match the client event log, and if the public key is in the unlocked status. The third-party server can then proceed with the authentication process. The third-party server grants access to the user account during Step M, if the validation token is received by the third-party server. In further detail, access is granted because the public credential input matches the public key and the collected user metadata corresponds to the user profile by using the client event log. Therefore, the public credential input passed the authentication process by the network of computer nodes, and the public key is validated to be in the unlocked status.

Alternatively, and with reference to FIG. 14, the validation token is also relayed with the remote server to the third-party server after Step L, if the public credential input does not match the public key or if the user metadata does not match the client event log, and if the public key is in the unlocked status. In this situation, the user metadata analysis results in inconsistencies or the public credential input does not match the public key. However, since the public key is in the unlocked status, the third-party server can then proceed with the authentication process. The third-party server denies access to the user account during Step N, if the validation token is received by the third-party server, due to the public credential input not matching the public key or the user metadata not matching the client event log. In further detail, access is denied because the public credential input does not match the public key, and therefore, the public credential input failed the authentication preprocess, and the public key was validated to be in the unlocked status. The service provider ensures this subprocess is legitimate by comparing the user metadata with the client event log. Furthermore, to keep the accuracy of the client event log up to date and to facilitate future automatic detection of fraudulent activities, and with reference to FIG. 15, the third-party server generates an access denial report from the user metadata after Step N, if the user account is denied access to the third-party server. The access denial report includes the user metadata relevant to the denied access that can be analyzed for future automatic detections of fraudulent activities. The access denial report is relayed from the third-party server to the remote server for processing and storage. The access denial report is also recorded and verified with the blockchain ledger for greater security. From the remote server, the relevant parties can access the access denial report as necessary.

In order for the remote server to invalidate the credential set before authentication is processed by the third-party server, and with reference to FIG. 16, the following subprocess is executed. Before the third-party server can authenticate the public credential input provided by the client account, the third-party server requests a token from the remote server. An invalidation token is relayed with the remote server to the third-party server after Step L, if the public credential input does match or does not match the public key, and if the public key is in the locked status. Since the credential set is in the locked status, the third-party server does not proceed with authentication process. The third-party server invalidates access to the user account during Step O, if the invalidation token is received by the third-party server. In further detail, the user account is prevented from being accessed altogether because the public key is in the locked status. This prevents access to the client account even if the public credential input matches the public key and if the user metadata corresponds to the client event log. Furthermore, to keep the accuracy of the client event log up to date and to facilitate future automatic detection of fraudulent activities, and with reference to FIG. 17, the third-party server generates an access invalidation report from the user metadata after Step O, if the user account is invalidated access to the third-party server. The access invalidation report includes the user metadata relevant to the invalidated access that can be analyzed for future automatic detections of fraudulent activities. The access invalidation report is relayed from the third-party server to the remote server for processing and storage. The access invalidation report is also recorded and verified with

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the blockchain ledger for greater security. From the remote server, the relevant parties can access the access invalidation report as necessary.

As previously discussed, the present invention facilitates the automatic detection of fraudulent activities during the account protection check so that the service provider can provide greater security to the client accounts. In reference to FIG. 18, to do so, the present invention is further provided with at least one policy guideline managed by the third-party server (Step Q). The policy guideline includes one or more set of rules that determine when the service provider must automatically block access the client account. Further, the client event log may include a plurality of log entries. Each log entry of the plurality of log entries corresponds to the various compilations of user metadata that occur when the user tries accessing the client account. The policy guideline provides policy management features that enable the service provider to predefine the parameters required to automatically block activity from detected fraudulent PC devices based on several parameters such as, but not limited to, Internet Protocol (IP) address, IP address ranges, varying timeframes, varying geographic locations, specific user agents, specific software applications, and/or specific user/client accounts. The subprocess for facilitating the automatic detection of fraudulent activities starts by comparing each log entry to the policy guideline with the third-party server before Step D in order to identify at least one guideline-inappropriate entry from the plurality of log entries (Step R). This enables the service provider to automatically detect any possible fraudulent activities by comparing the user metadata with the previous collected metadata in the plurality of log entries. Then, the third-party server generates a fraudulent activity report from the guideline-inappropriate entry, if the guideline-inappropriate entry is identified from the plurality of log entries (Step S). The fraudulent activity report can be a compilation of data related to the different guideline-inappropriate entries identified during Step T. The fraudulent activity report is relayed from the third-party server to the remote server (Step M) for storage and processing. Then, to protect the client account, the remote server automatically sets the public key to the locked status, if the guideline-inappropriate entry is identified from the plurality of log entries, and if the public key is in the unlocked status (Step U). This way, the service provider is able to automatically detect potentially fraudulent activities and improve account protection services. For example, the service provider can return list and details of the identified fraudulent activities to external services, as well as to recommend actions to take based on the same parameters defined in the policy guidelines. For example, the service provider may call a firewall management Application Programming Interface (API) to create firewall rules dynamically, call another policy management system to create/modify a locking/enabling policy, or call intrusion detection systems to trigger for increased levels of activity monitoring.

In addition to automatically detect fraudulent activities, the present invention is able to improve the automatic detection capabilities through machine learning. In reference to FIG. 19, to do so, each event log includes log characteristic information that can be used to determine potential new fraudulent behavior not previously detected by the present invention. Then, the third-party server modifies the appropriate policy guideline in accordance to log characteristic information of the guideline-inappropriate entry. In other words, the policy guideline is modified when there is some kind of acute fraudulent activity. For example, a specific IP

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address from a foreign country may be banned if fraudulent activity is performed using said IP address. The policy guideline would be updated to block any activity from the specific IP address.

In reference to FIG. 20, this subprocess may be repeated continuously so that automatic detection capabilities are updated to automatically detect trends of fraudulent activity. To do so, a plurality of iterations for Steps R through U is executed in order to identify at least one new fraudulent-activity trend from the fraudulent activity report of each iteration for Steps R through U. Then, the third-party server modifies the policy guideline in accordance to the new fraudulent-activity trend anytime a new fraudulent behavior is detected in the log characteristic information. The continuous detection of fraudulent activities enables the reporting and analysis of PC devices, users, and geolocation activity from which potentially fraudulent activity is coming from over time. For example, if different IP addresses are detected to perform fraudulent activities and all those IP addresses come from a same geographical region or country, the policy guideline would be updated to block any activity coming from the geographical region or country. The service provider can generate maps of locations/networks, list of PC devices, networks, software applications making requests, as well as geographic locations from which the fraudulent activity is coming from. Then, the service provider or external services may be able to filter any/all these fraudulent activity parameters, including time.

Exemplary Detection Algorithms

The present invention may utilize several algorithms/patterns to determine if activity is fraudulent in nature relying on the status of the public key and the client event log. If multiple authentications originate from a single IP, a network, or a geolocation, and are made against multiple unique public keys which are locked, this is an indication that the actor making the requests is attempting to authenticate public keys not owned by them. For example, a first IP address makes authentication attempts against account A which is locked, account B which is locked, and account C which is locked over the course of two minutes (within a configurable time window). The number of failed attempts based on the lock status and the time frame over which the authentication attempts occur are configurable. Another example, requests to authenticate a public key/account come from a first IP address. If the account was locked/unlocked in the previous two minutes from a second IP address, this means that the locking or the unlocking attempts are not coming from the same host as the one attempting to authenticate. This is also an indication of potentially fraudulent activity.

Another detection algorithm involves account locking/unlocking calls which can be restricted to be allowed only from specific status constraints given IP address, IP Ranges, geographic locations, and during specific times. For example, only the supplied IP address is allowed to lock/unlock a given public key. If an IP address, or range of unique IP addresses, make requests to lock/unlock public keys against a range of accounts not previously associated with the account, then this is an indication that the lock/unlock activity is potentially fraudulent and future lock/unlock (and all other requests into other systems also) can be blocked or monitored with a finer degree of event logging based on policies or rules.

Another detection algorithm involves utilizing machine learning and Artificial Intelligence (AI). The service provider can apply probabilistic classifiers to automatically compute statistical matches classifying lock/unlock related

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activity as potentially fraudulent. For example, the service provider can leverage Bayesian statistics to automatically compare known successful lock/unlock and authentication patterns for credential sets against known failed attempts for a given IP, IP range, geographical location, or public key. If an account is unlocked from a given IP address and authentication using a given public key is successful, the service provider can infer that the user is validated successfully. If, however, a different IP address makes a request unsuccessfully against an unlocked/locked account or makes multiple unsuccessful authentication attempts (credentials supplied are invalid), then the service provider can infer that the user is invalid and block activity.

Another detection algorithm involves utilizing PC device characteristics and identifiers so that the service provider can register approved devices for each account. Unlock attempts from other devices can require additional levels of authentication. For example, a PC device that has never been used to unlock an account can cause an email or text message to be sent to the email or cell phone on file with a code that must be entered or a link that must be clicked to validate that the new PC device is trusted. For highly sensitive or shared accounts, policy guidelines can be set by the user or by the organization to which the user belongs to require an N number of approvals to unlock an account. These approvals can be required either from N number of people in the organization, users within a specific group, or specific named individuals. For example, to unlock certain administrative accounts in an enterprise, the organization can apply a policy that requires N number of approvals before the account can be unlocked. For a shared bank account, both parties would have to provide approval for the account to be unlocked.

Another detection algorithm involves utilizing thresholds that can be set for unlocked accounts which only allow the public keys to be used an N number of times before automatically locking again. These thresholds can be chosen by the end user or enforced via policy guidelines. Policy can specify any protection check or specify failed or successful if that information is being provided to the service. For example, upon unlocking an account, a user can choose the number of times the user expects to use the public key. If the expectation is one time and the user sets the threshold to one, then the public key will automatically relock after the next account protection check. Another detection algorithm involves users or organizations able to choose to enforce a geography check on the account when the account is unlocked by setting the allowed geographic location(s) for the account or by setting a geography during the unlock process. This would prevent bad actors from using the public keys from other geographies while the accounts are unlocked.

Further, if information about a failed login attempt after protection check is being provided, the policy guideline can be set to automatically re-lock the account after a failed login. This would provide enhanced protection but would require users to unlock the account again after a legitimate failed login. This policy guideline can be set per public key, for groups of public keys, or all public keys. Further, if certain credentials are only meant to be used in specific locations, and if the organization can provide details from their access control system on when users have badged in and out of the location, then the public keys can be restricted from being unlocked if the user is not badged into the building. Alternatively, the policy can be set to automatically lock and unlock the accounts when the users enter and exit the location. Further, the policy guideline can be set to

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require multi-function authentication to lock or unlock specific public keys or groups of public keys.

Although the invention has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of the invention as hereinafter claimed.

What is claimed is:

1. A method for facilitating an account protection check through blockchain, the method comprising the steps of:
 - (A) providing at least one client account managed by at least one remote server, wherein the client account is associated with a corresponding personal computing (PC) device;
 - (B) providing at least one user account managed by at least one third-party server, wherein the user account is associated with the corresponding PC device, and wherein the user account is associated to the client account;
 - (C) providing at least one public key managed by at least one blockchain ledger, wherein the public key is a first piece for a set of credentials, and wherein the public key is associated with the client account and the user account, and wherein the public key is either in an unlocked status or a locked status, and wherein the blockchain ledger is managed by a network of computing nodes;
 - (D) prompting the client account to enter a public credential input for the user account with the corresponding PC device;
 - (E) relaying the public credential input from the corresponding PC device, through the third-party server, through the remote server, and to the network of computing nodes, if the public credential input is entered by the client account;
 - (F) granting access to the user account with the third-party server, if the public credential input does match the public key, and if the public key is in the unlocked status;
 - (G) denying access to the user account with the third-party server, if the public credential input does not match the public key, and if the public key is in the unlocked status;
 - (H) invalidating access to the user account with the third-party server, if the public credential input does match or does not match the public key, and if the public key is in the locked status; and
 - (I) recording and verifying steps (F), (G), or (H) with the blockchain ledger.
2. The method as claimed in claim 1 comprising the steps of:
 - providing at least one private key managed by the blockchain ledger, wherein the private key is a second piece for the set of credentials, and wherein the private key is associated with the client account and the user account;
 - prompting the client account to enter a private credential input for the user account with the corresponding PC device;
 - prompting the client account to link to the user account with the corresponding PC device during step (B), if the private credential input does match the private key; and
 - generating an association between the user account and the client account with the remote server, if the user account is selected to be linked to the client account.
3. The method as claimed in claim 1 comprising the steps of:

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providing at least one private key managed by the block-chain ledger, wherein the private key is a second piece for the set of credentials, and wherein the private key is associated with the client account and the user account;

prompting the client account to enter a private credential input for the user account with the corresponding PC device;

prompting the client account to enter a single status selection for the public key with the corresponding PC device, if the private credential input does match the private key;

relaying the single status selection and the private credential input from the corresponding PC device to the remote server, if the single status selection is entered by the client account;

setting the public key to the unlocked status with the remote server, if the single status selection is for the unlocked status; and

setting the public key to the locked status with the remote server, if the single status selection is for the locked status.

4. The method as claimed in claim 1 comprising the steps of:

providing at least one private key managed by the block-chain ledger, wherein the private key is a second piece for the set of credentials, and wherein the private key is associated with the client account and the user account;

providing at least one user account as a plurality of user accounts;

prompting the client account to enter a private credential input for the user account with the corresponding PC device;

prompting the client account to enter a group status selection for the public key of each user account with the corresponding PC device, if the private credential input does match the private key;

relaying the group status selection from the corresponding PC device to the remote server, if the group status selection is entered by the client account;

setting the public key to the unlocked status with the remote server, if the group status selection is for the unlocked status; and

setting the public key to the locked status with the remote server, if the group status selection is for the locked status.

5. The method as claimed in claim 1 comprising the steps of:

providing the public key in the unlocked status;

providing at least one private key managed by the block-chain ledger, wherein the private key is a second piece for the set of credentials, and wherein the private key is associated with the client account and the user account;

prompting the client account to enter a private credential input for the user account with the corresponding PC device;

prompting the client account to enter at least one status constraint for the unlocked status with the corresponding PC device, if the private credential input does match the private key;

relaying the status constraint from the corresponding PC device to the remote server, if the status constraint is entered by the client account;

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maintaining the public key in the unlocked status with the remote server, if the unlocked status is inside the status constraint; and

resetting the public key from the unlocked status to the locked status with the remote server, if the unlocked status is outside the status constraint.

6. The method as claimed in claim 1 comprising the steps of:

providing the public key in the locked status;

providing at least one private key managed by the block-chain ledger, wherein the private key is a second piece for the set of credentials, and wherein the private key is associated with the client account and the user account;

prompting the client account to enter a private credential input for the user account with the corresponding PC device;

prompting the client account to enter at least one status constraint for the locked status with the corresponding PC device, if the private credential input does match the private key;

relaying the status constraint from the corresponding PC device to the remote server, if the status constraint is entered by the client account;

maintaining the public key in the locked status with the remote server, if the locked status is inside the status constraint; and

resetting the public key from the locked status to the unlocked status with the remote server, if the locked status is outside the status constraint.

7. The method as claimed in claim 1 comprising the steps of:

relaying a validation token with the remote server to the third-party server after step (E), if the public credential input does match the public key, and if the public key is in the unlocked status; and

granting access to the user account with the third-party server during step (F), if the validation token is received by the third-party server.

8. The method as claimed in claim 1 comprising the steps of:

relaying a validation token with the remote server to the third-party server after step (E), if the public credential input does not match the public key, and if the public key is in the unlocked status; and

denying access to the user account with the third-party server during step (G), if the validation token is received by the third-party server.

9. The method as claimed in claim 1 comprising the steps of:

relaying an invalidation token with the remote server to the third-party server after step (E), if the public credential input does match or does not match the public key, and if the public key is in the locked status; and

invalidating access to the user account with the third-party server during step (H), if the invalidation token is received by the third-party server.

10. The method as claimed in claim 1 comprising the steps of:

generating an invalid notification with the third-party server during step (H), if the public credential input does match or does not match the public key, and if the public key is in the locked status; and

displaying the invalid notification with the corresponding PC device after step (H).

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11. The method as claimed in claim 1 comprising the steps of:

- (J) providing the client account with a client event log, wherein the client event log is managed by the remote server;
- (K) providing the user account with user metadata, wherein the user metadata is managed by the third-party server;
- (L) relaying the user metadata from the corresponding PC device, through the third-party server, through the remote server, and to the network of computing nodes during step (E);
- (M) executing step (F), if the public credential input does match the public key, and if the user metadata does match the client event log, and if the public key is in the unlocked status;
- (N) executing step (G), if the public credential input does not match the public key or if the client metadata does not match the client event log, and if the public key is in the unlocked status;
- (O) executing step (H), if the public credential input does match or does not match the public key, and if the public key is in the locked status; and
- (P) recording and verifying the client event log and the user metadata with the blockchain ledger.

12. The method as claimed in claim 11 comprising the steps of:

- relaying a validation token with the remote server to the third-party server after step (L), if the public credential input does match the public key, and if the user metadata does match the client event log, and if the public key is in the unlocked status; and
- granting access to the user account with the third-party server during step (M), if the validation token is received by the third-party server.

13. The method as claimed in claim 11 comprising the steps of:

- relaying a validation token with the remote server to the third-party server after step (L), if the public credential input does not match the public key or if the user metadata does not match the client event log, and if the public key is in the unlocked status; and
- denying access to the user account with the third-party server during step (N), if the validation token is received by the third-party server.

14. The method as claimed in claim 13 comprising the steps of:

- generating an access denial report from the user metadata with the third-party server after step (N), if the user account is denied access to the third-party server;
- relaying the access denial report from the third-party server to the remote server; and
- recording and verifying the access denial report with the blockchain ledger.

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15. The method as claimed in claim 11 comprising the steps of:

- relaying an invalidation token with the remote server to the third-party server after step (L), if the public credential input does match or does not match the public key, and if the public key is in the locked status; and
- invalidating access to the user account with the third-party server during step (O), if the invalidation token is received by the third-party server.

16. The method as claimed in claim 15 comprising the steps of:

- generating an access invalidation report from the user metadata with the third-party server after step (O), if the user account is invalidated access to the third-party server;
- relaying the access invalidation report from the third-party server to the remote server; and
- recording and verifying the access invalidation report with the blockchain ledger.

17. The method as claimed in claim 11 comprising the steps of:

- (Q) providing at least one policy guideline managed by the third-party server, wherein the client event log includes a plurality of log entries;
- (R) comparing each log entry to the policy guideline with the third-party server before step (D) in order to identify at least one guideline-inappropriate entry from the plurality of log entries;
- (S) generating a fraudulent activity report from the guideline-inappropriate entry with the third-party server, if the guideline-inappropriate entry is identified from the plurality of log entries;
- (T) relaying the fraudulent activity report from the third-party server to the remote server; and
- (U) setting the public key to the locked status with the remote server, if the guideline-inappropriate entry is identified from the plurality of log entries, and if the public key is in the unlocked status.

18. The method as claimed in claim 17 comprising the steps of:

- providing each event log with log characteristic information; and
- modifying the policy guideline in accordance to log characteristic information of the guideline-inappropriate entry with the third-party server.

19. The method as claimed in claim 17 comprising the steps of:

- executing a plurality of iterations for steps (R) through (U) in order to identify at least one new fraudulent-activity trend from the fraudulent activity report of each iteration for steps (R) through (U); and
- modifying the policy guideline in accordance to the new fraudulent-activity trend with the third-party server.

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